Notes of 104016 Algebra 1/Extended

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1 Some abbreviations and notations

1.1 Abbreviations

Throughout this notes, we will use the following abbreviations:

- s.t. : such that
- coeff. : coefficient
- iff : if and only if
- rep. : representation
- l.s. : linear system
- l.i. : linear independent
- l.d. : linear dependent
- l.c. : linear combination
- l.o. : linear operator
- l.t. : linear transformation
- v.s. : vector space
- f.d.v.s. : finite dimensional vector space
- n-d.v.s. : n-dimensional vector space
- alt. : alternating
- mult. : multilinear

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1.2 Notations

Throughout this notes, we will use the following notations:

- $\mathbb{N}$: the set of natural numbers $\{1, 2, 3, \ldots\}$
- $\mathbb{N}_0$: the set of natural numbers including zero $\{0, 1, 2, 3, \ldots\}$
- $\mathbb{Z}$: the set of integers $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$
- $\mathbb{Q}$: the set of rational numbers
- $\mathbb{R}$: the set of real numbers
- $\mathbb{C}$: the set of complex numbers
- $\mathbb{F}$: a general field
- $\mathbb{F}_q$: a finite field with q elements
- $\mathbb{Z}_p$ or $\mathbb{F}_p$: the finite field of integers modulo a prime number p

2 Fields, finite fields and modular arithmetic

2.1 Fields

2.1.1 Definition of a field

Formally, a field is a set $\mathbb{F}$ together with two binary operations on $\mathbb{F}$ called addition and multiplication. And it satisfies the following axioms (for all $x, y, z \in \mathbb{F}$):

For additive operation:

A1 Commutativity: $x + y = y + x$

A2 Associativity: $x + (y + z) = (x + y) + z$

A3 Identity: There is a unique element 0 (zero) in $\mathbb{F}$ such that $x + 0 = x$, for every x in $\mathbb{F}$.

A4 Inverse: To each x in $\mathbb{F}$, there corresponds a unique element $-x$ in $\mathbb{F}$ such that $x + (-x) = 0$.

For multiplicative operation:

M1 Commutativity: $xy = yx$

M2 Associativity: $x(yz) = (xy)z$

M3 Identity: There is a unique non-zero element 1 (one) in $\mathbb{F}$ such that $x1 = x$, for every x in $\mathbb{F}$.

M4 Inverse: To each non-zero x in $\mathbb{F}$ there corresponds a unique element x^{-1} in $\mathbb{F}$ such that $xx^{-1} = 1$.

Also, we have multiplication distributes over addition:

D Distributivity: $x(y + z) = xy + xz$

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2.2 Finite fields

2.2.1 Definition of finite fields

A finite field (or Galois field) is a field that has a finite number of elements. As with any field, a finite field is a set on which the operations of addition and multiplication are defined and satisfy certain basic rules. The most common examples of finite fields are the integers mod p when p is a prime number.

2.2.2 A example of finite field

Question:

Let $\mathbb{F}_3$ be the set $0, 1, 2$ furnished with the following binary operations: $" + " := +(\text{mod } 3)$ and $" \cdot " := \cdot(\text{mod } 3)$

1. Write the addition and multiplication tables for $\mathbb{F}_3$.
2. Prove that $\mathbb{F}_3$ is a field.

Solution:

1. Addition table:

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

Multiplication table:

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

2. It's easy to prove that all the field axioms are satisfied by checking the addition and multiplication tables above. So the proof will be omitted here.

2.2.3 The finite fields Z_p

For any prime number p, the set of integers modulo p, denoted by $\mathbb{Z}_p$ or $\mathbb{F}_p$, forms a finite field with p elements under addition and multiplication modulo p. The elements of $\mathbb{Z}_p$ are $\{0, 1, 2, \ldots, p-1\}$.

In $\mathbb{Z}_p$, addition and multiplication are defined as follows:

- Addition: For any two elements $a, b \in \mathbb{Z}_p$, their sum is given by $(a + b) \mod p$.
- Multiplication: For any two elements $a, b \in \mathbb{Z}_p$, their product is given by $(a \cdot b) \mod p$.

The field axioms can be verified for $\mathbb{Z}_p$, making it a valid finite field.

2.3 Modular arithmetic

2.3.1 Definition of modular arithmetic

Modular arithmetic is a system of arithmetic operations for integers, other than the usual ones from elementary arithmetic, where numbers "wrap around" when reaching a certain value, called the modulus.

2.3.2 Notation of modular arithmetic

In modular arithmetic, we say that two integers a and b are congruent modulo n if they have the same remainder when divided by n. This is written as:

$$a \equiv b \pmod{n} \tag{1}$$

This means that n divides the difference of a and b, or equivalently, there exists an integer k such that:

$$a - b = kn \tag{2}$$

For example, $17 \equiv 5 \pmod{12}$ because both 17 and 5 leave a remainder of 5 when divided by 12, and $17 - 5 = 12$ which is a multiple of 12.

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3 Rational, real and complex numbers

3.1 Rational numbers

3.1.1 Definition of rational numbers

A rational number is any number that can be expressed as the quotient or fraction $\frac{p}{q}$ of two integers, where the numerator p is an integer and the denominator q is a non-zero integer. The set of all rational numbers is denoted by $\mathbb{Q}$.

3.2 Real numbers

3.2.1 Definition of real numbers

Real numbers include all the rational numbers, such as integers and fractions, as well as all the irrational numbers, which cannot be expressed as fractions. The set of real numbers is denoted by $\mathbb{R}$.

3.3 Complex numbers

3.3.1 Definition of complex numbers

A complex number is a number that can be expressed in the form $a + bi$, where a and b are real numbers, and i is the imaginary unit, which satisfies the equation $i^2 = -1$. The set of all complex numbers is denoted by $\mathbb{C}$.

a is called the real part of the complex number, and b is called the imaginary part. The notation is $a = \mathrm{Re}$, $b = \mathrm{Im}$.

3.3.2 Notations

The conjugate of a complex number $z = a + bi$ is denoted by $\overline{z}$ and is defined as:

$$\overline{z} = a - bi \tag{3}$$

The modulus (or absolute value) of a complex number $z = a + bi$ is denoted by $|z|$ and is defined as:

$$|z| = \sqrt{a^2 + b^2} \tag{4}$$

3.3.3 Operations and properties

Denote two complex numbers as $z_1 = a + bi$ and $z_2 = c + di$, where a, b, c, d are real numbers. The basic operations on complex numbers are defined as follows:

- Addition: $z_1 + z_2 = (a + c) + (b + d)i$
- Subtraction: $z_1 - z_2 = (a - c) + (b - d)i$
- Multiplication: $z_1 \cdot z_2 = (ac - bd) + (ad + bc)i$
- Division: $\frac{z_1}{z_2} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$, provided that $z_2 \neq 0$

Therefore, we have the following properties:

- Conjugate of a sum: $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$
- Conjugate of a product: $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$
- Modulus of a product: $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$
- Modulus of a quotient: $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$, provided that $z_2 \neq 0$

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3.4 Euler's formula and De Moivre's Theorem

3.4.1 Statement of Euler's Formula

Euler's formula states that for any real number θ:

$$e^{i\theta} = \cos(\theta) + i\sin(\theta) \tag{5}$$

Or, we can denote $e^{i\theta}$ as $\text{cis}(\theta)$:

$$\text{cis}(\theta) = \cos(\theta) + i\sin \tag{6}$$

Where θ is the argument of the complex number, denoted as $\arg(z)$.

3.4.2 Statement of De Moivre's Theorem

De Moivre's Theorem states that for any real number θ and integer n:

$$(\cos(\theta) + i\sin(\theta))^n = \cos(n\theta) + i\sin(n\theta) \tag{7}$$

This theorem is useful for finding powers and roots of complex numbers.

3.4.3 Applications

Denote two complex number z_1 and z_2 in polar form: $z_1 = r_1\text{cis}(\theta_1)$ and $z_2 = r_2\text{cis}(\theta_2)$, where r_1, r_2 are the moduli and θ_1, θ_2 are the arguments of z_1 and z_2 respectively. Then we have:

- Multiplication: $z_1 \cdot z_2 = r_1 r_2 \text{cis}(\theta_1 + \theta_2)$
- Division: $\frac{z_1}{z_2} = \frac{r_1}{r_2}\text{cis}(\theta_1 - \theta_2)$, provided that $z_2 \neq 0$
- Power: $z_1^n = r_1^n \text{cis}(n\theta_1)$
- Inverse: $z_1^{-1} = \frac{1}{r_1}\text{cis}(-\theta_1)$, provided that $z_1 \neq 0$
- Conjugate: $\overline{z_1} = r_1\text{cis}(-\theta_1)$

4 Polynomials

4.1 Definition of polynomials

Let $\mathbb{F}$ be a field and consider a indeterminate x. A polynomial in the variable x with cofficients in $\mathbb{F}$ (or a polynomial in $\mathbb{F}[x]$) is an expression of the form:

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0 \tag{8}$$

Where $a_0, a_1, \ldots, a_n \in \mathbb{F}$ are the cofficients, and n is in $\mathbb{N}_0$. The set of all such polynomials is denoted by $\mathbb{F}[x]$.

4.2 Notations and definitions over polynomials

In the polynomial $p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$, the following notations and definitions are used:

- Degree: The degree of a non-zero polynomial $p(x)$, denoted by $\deg(p)$, is the largest integer n s.t. $a_n \neq 0$.
- Leading term: The leading term is the term $a_n x^n$ where $n = \deg(p)$.
- Leading coefficient: The leading coefficient is the coefficient a_n of the leading term.
- Costant coefficient: The constant coeff. a_0.
- Monic polynomial: A polynomial is called <u>monic</u> if the leading coeff. is 1.
- Zero polynomial: $\deg(0) = -\infty$.
- Constant polynomial: A polynomial $p(x)$ is called constant if $\deg(p) = 0$, and $p(x) = a_0$.
- By convention, $x^0 = 1$.

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4.3 Polynomial division

4.3.1 Dividible polynomials

Let $p(x), d(x) \in \mathbb{F}[x]$ be two polynomials, with $d(x) \neq 0$. We say that $d(x)$ divides $p(x)$ (denoted by $d(x)|p(x)$) if there exists a polynomial $q(x) \in \mathbb{F}[x]$ such that:

$$p(x) = d(x)q(x) \tag{9}$$

In this case, $q(x)$ is called the quotient polynomial.

4.3.2 Some properties of divisibility

Let $f(x)$, $g(x)$, $h(x) \in \mathbb{F}[x]$ and $c \in \mathbb{F}\backslash\{0\}$. Then we have the following properties:

- $f(x)|0$
- $c|f(x)$
- $cf(x)|f(x)$
- if $f(x)|g(x)$ and $g(x)|h(x)$, then $f(x)|h(x)$
- if $f(x)|g(x)$, then $f(x)|g(x)h(x)$
- if $f(x)|g(x)$ and $f(x)|h(x)$, then $f(x)|(g(x) + h(x))$
- if $f(x)|c$, then $\deg(f(x)) = 0$
- if $f(x)|g(x)$ and $g(x)|f(x)$, then $\exists d \in \mathbb{F}\backslash\{0\}$ s.t. $f(x) = dg(x)$
- $f(x)|g(x)$ iff $cf(x)|g(x)$ iff $f(x)|cg(x)$

4.4 Theorem of common divisor

4.4.1 Statement of the theorem

Let $f(x)$ and $g(x)$ be polynomials in $\mathbb{F}[x]$, not both zero. Then there exists a unique monic polynomial $h(x)$ of highest degree dividing both $f(x)$ and $g(x)$. Moreover, any polynomial that divides $f(x)$ and $g(x)$ also divides $h(x)$.

Furthermore, there exist polynomials $r(x)$ and $t(x)$ s.t.:

$$h(x) = r(x)f(x) + t(x)g(x) \tag{10}$$

The polynomial $h(x)$ is called the greatest common divisor (gcd) of $f(x)$ and $g(x)$:

$$\text{g.c.d.}(f(x), g(x)) = h(x) \tag{11}$$

We say that $f(x)$ and $g(x)$ are coprime if g.c.d.$(f(x), g(x)) = 1$.

4.4.2 Lemma of this theorem

If $f(x)$, $g(x)$, $h(x) \in \mathbb{F}[x]$ and $f(x)|g(x)h(x)$ with g.c.d.$(f(x), g(x)) = 1$, then $f(x)|h(x)$.
Proof see section "Proofs".

4.4.3 Extension of this theorem

If $f(x)|h(x)$ and $g(x)|h(x)$ where g.c.d.$(f(x), g(x)) = 1$, then $f(x)g(x)|h(x)$.

4.5 Fundamental theorem of arithmetics

4.5.1 Statement of the theorem

Given $f(x) \in \mathbb{F}[x]$, $f(x) \neq 0$ and $\deg(f(x)) \geq 1$. Then there exist $p_1(x), p_2(x), \cdots, p_r(x) \in \mathbb{F}[x]$ irreducible and monic, $c \in \mathbb{F}\backslash\{0\}$ s.t.:

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$$f(x) = c(p_1(x))^{n_1}(p_2(x))^{n_2}\cdots(p_r(x))^{n_r} = \prod_{i=1}^{r}(p_i(x))^{n_i} \tag{12}$$

4.5.2 Remark

$p_1(x), p_2(x), \cdots, p_r(x)$ are unique up to permutation.
$n_1, n_2, \cdots, n_r$ are unique up to permutation.
c is unique.

4.6 Fundamental theorem of algebra

4.6.1 Statement of the theorem

$\mathbb{C}$ is algebraically closed $\equiv$ every non-constant polynomial in $\mathbb{C}[x]$ has at least one root in $\mathbb{C}$.

4.7 Theorems and algorithms about polynomial division and roots

4.7.1 Long division of polynomials

Let $p(x)$, $q(x) \in \mathbb{F}[x]$, then $\exists c(x)$, $r(x) \in \mathbb{F}[x]$ s.t.:

$$p(x) = c(x)q(x) + r(x) \tag{13}$$

Where $\deg(r(x)) < \deg(q(x))$ or $r(x) = 0$.

4.7.2 Remainder theorem

Let $p(x) \in \mathbb{F}[x]$, and $\deg(p(x)) \geq 1$. Let $\alpha \in \mathbb{F}$, then $\exists c(x) \in \mathbb{F}[x]$ s.t.:

$$p(x) = (x - \alpha)c(x) + p(\alpha) \tag{14}$$

4.7.3 Factor theorem

Let $p(x) \in \mathbb{F}[x] \backslash \{0\}$, and $\deg(p(x)) \geq 1$. Let $\alpha \in \mathbb{F}$, then $(x-\alpha)|p(x)$ iff $p(\alpha) = 0$.

4.7.4 Gauss' criterion

Given $f(x) \in \mathbb{Z}[x]$, $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$, $a_n \neq 0$ of degree n. Let $r \in \mathbb{Z}$, $s \in \mathbb{N}$ s.t. g.c.d.$(r, s) = 1$ and $\frac{r}{s}$ is a root of $f(x)$ in $\mathbb{Q}$. Then $s|a_n$ and $r|a_0$.

4.7.5 Eisentein's criterion

Let $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a monic polynomial in $\mathbb{Z}[x]$. If p is a prime number s.t.: $p|a_0$, a_1, $\cdots$, a_{n-1}, but $p^2 \nmid a_0$, then $f(x)$ is irreducible in $\mathbb{Q}[x]$.

4.7.6 Vieta's formular

Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = a_n(x-x_1)(x-x_2)\cdots(x-x_n)$
Then we have:

$$\sum_{i=1}^{n} x_i = -\frac{a_{n-1}}{a_n} \tag{15}$$

$$\sum_{1 \leq i < j \leq n} x_i x_j = \frac{a_{n-2}}{a_n} \tag{16}$$

$$\sum_{1 \leq i < j < k \leq n} x_i x_j x_k = -\frac{a_{n-3}}{a_n} \tag{17}$$

$$\cdots$$

$$\prod_{i=1}^{n} x_i = (-1)^n \frac{a_0}{a_n} \tag{18}$$

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5 Vectors

5.1 Vectors in $\mathbb{R}^2$

5.1.1 Definition of vectors in $\mathbb{R}^2$

We define $\mathbb{R}^2$ as the set of pairs (a, b) with $a, b \in \mathbb{R}$. $\mathbb{R}^2 = \{(a, b) | a, b \in \mathbb{R}\}$
A vector in the plane is a directed segment $\overrightarrow{PQ}$ from an initial point P to a final point Q.
Each vector has:

- A line action, the line passing through P and Q.
- An direction, from P to Q.
- A magnitude (or length), denoted by $|\overrightarrow{PQ}|$.

5.1.2 Equivalent vectors

Two vectors are equivalent if

i They are parallel.

ii They have the same direction.

iii They have the same magnitude.

Notice that any vector is equivalent to a unique vector whose initial point is $(0, 0) = O$.

5.1.3 Vector operation

Consider $\overrightarrow{v} = (a, b)$ and $\overrightarrow{w} = (c, d)$ in $\mathbb{R}^2$, we define:

- Addition: $\overrightarrow{v} + \overrightarrow{w} = (a + c, b + d)$
- Scalar multiplication: For any $k \in \mathbb{R}$, $k\overrightarrow{v} = (ka, kb)$

5.1.4 Line through a certain point

In general, given v_0 and v in $\mathbb{R}^2$, $l = \{v_0 + tV \mid t \in \mathbb{R}\}$ is the line through point v_0 in the direction of vector v.

5.2 Vectors in $\mathbb{R}^3$

5.2.1 A line in $\mathbb{R}^3$

A line in $\mathbb{R}^3$ is the set $\{(x, y, z) \in \mathbb{R}^3 \mid (x, y, z) = \overrightarrow{v} + t\overrightarrow{w}, t \in \mathbb{R}\}$

Normally, we write: $l : (x, y, z) = (a, b, c) + t(\alpha, \beta, \gamma)$, $t \in \mathbb{R}$, this is called the parametric rep. of the line.

Another rep. is $l : (x, y, z) = (a + t\alpha, b + t\beta, c + t\gamma)$, $t \in \mathbb{R} \Rightarrow t = \frac{x-a}{\alpha} = \frac{y-b}{\beta} = \frac{z-c}{\gamma}$. This is called the algebraic rep. of the line.

5.2.2 Mutual relations of two lines in $\mathbb{R}^3$

Given two lines l_1 and l_2 in $\mathbb{R}^3$:

i intersect (incident): $l_1 \cap l_2 \neq \emptyset$ and $\overrightarrow{v_1} \nparallel \overrightarrow{v_2}$

ii parallel: $l_1 \cap l_2 = \emptyset$ and $\overrightarrow{v_1} \parallel \overrightarrow{v_2}$

iii coincide: $l_1 = l_2$

iv skew: $l_1 \cap l_2 = \emptyset$ and $\overrightarrow{v_1} \nparallel \overrightarrow{v_2}$

5.2.3 Plane's equation

Let (a, b, c) be a point in space and let $\overrightarrow{N} = (A, B, C)$ be a vector. The set of all points $(x, y, z) \in \mathbb{R}^3$ s.t. $((x, y, z) - (a, b, c)) \perp \overrightarrow{N}$ is called a "plane".

Mathematically speaking,

$$\begin{aligned}\text{plane} &= \{(x, y, z) \in \mathbb{R}^3 \mid ((x, y, z) - (a, b, c)) \cdot \overrightarrow{N} = 0\} \\ &= \{(x, y, z) \in \mathbb{R}^3 \mid A(x - a) + B(y - b) + C(z - c) = 0\} \\ &= \{(x, y, z) \in \mathbb{R}^3 \mid Ax + By + Cz + (-aA - bB - cC) = 0\}\end{aligned}$$

Therefore, normally we write the plane's equation as:

$$\pi : Ax + By + Cz + D = 0 \tag{19}$$

Where $D = -aA - bB - cC$.

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5.3 Vectors in general $\mathbb{F}^n$

5.3.1 Definition of vectors in $\mathbb{F}^n$

Let $\mathbb{F}$ be a field and n a natural number. We define $\mathbb{F}^n$ as the set of n-tuples $(a_1, a_2, \cdots, a_n)$ with elements in $\mathbb{F}$. That is: $\mathbb{F}^n = \{(a_1, a_2, \cdots, a_n) \mid a_i \in \mathbb{F}, i = 1, 2, \cdots, n\}$. And we call it's elements vectors.

Notation: for linear equations, we use $\begin{bmatrix} a_1 \\ a_2 \\ \cdots \\ a_n \end{bmatrix}$ for the vector $(a_1, a_2, \cdots, a_n)$.

5.3.2 Vector operations in $\mathbb{F}^n$

Consider $v = (a_1, a_2, \cdots, a_n)$ and $w = (b_1, b_2, \cdots, b_n)$ in $\mathbb{F}^n$, we define:

- Addition: $v + w = (a_1 + b_1, a_2 + b_2, \cdots, a_n + b_n)$
- Scalar multiplication: For any $c \in \mathbb{F}$, $cv = (ca_1, ca_2, \cdots, ca_n)$
- Zero vector: $0 = (0, 0, \cdots, 0)$
- Inverse vector: For any $v = (a_1, a_2, \cdots, a_n)$, $-v = (-a_1, -a_2, \cdots, -a_n)$

5.3.3 Linear combination

v in $\mathbb{R}^n$ is a linear combination of v_1, v_2, $\cdots$, v_k in $\mathbb{R}^n$ if there exist c_1, c_2, $\cdots$, c_k in $\mathbb{R}$ s.t.:

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k \tag{20}$$

5.3.4 Vector space axioms

$(\mathbb{F}^n, +, \cdot)$ satisfies the following axioms:

A1 Commutativity of addition: $\overrightarrow{u} + \overrightarrow{v} = \overrightarrow{v} + \overrightarrow{u}$

A2 Associativity of addition: $(\overrightarrow{u} + \overrightarrow{v}) + \overrightarrow{w} = \overrightarrow{u} + (\overrightarrow{v} + \overrightarrow{w})$

A3 Existence of additive identity: There exists a vector $\overrightarrow{0}$ s.t. $\overrightarrow{v} + \overrightarrow{0} = \overrightarrow{v}$ for all $\overrightarrow{v}$

A4 Existence of additive inverse: For each vector $\overrightarrow{v}$, there exists a vector $-\overrightarrow{v}$ s.t. $\overrightarrow{v} + (-\overrightarrow{v}) = \overrightarrow{0}$

P1 Compatibility of scalar multiplication with field multiplication: $a(b\overrightarrow{v}) = (ab)\overrightarrow{v}$ for all $a, b \in \mathbb{R}$ and all vectors $\overrightarrow{v}$

P2 Identity element of scalar multiplication: $1\overrightarrow{v} = \overrightarrow{v}$ for all vectors $\overrightarrow{v}$

P3 Distributivity of scalar multiplication with respect to vector addition: $a(\overrightarrow{u} + \overrightarrow{v}) = a\overrightarrow{u} + a\overrightarrow{v}$ for all $a \in \mathbb{R}$ and all vectors $\overrightarrow{u}$, $\overrightarrow{v}$

P4 Distributivity of scalar multiplication with respect to field addition: $(a + b)\overrightarrow{v} = a\overrightarrow{v} + b\overrightarrow{v}$ for all $a, b \in \mathbb{R}$ and all vectors $\overrightarrow{v}$

5.4 Remark

5.4.1 0 is in the span

$0 \in \text{span}\{v_1, v_2, \cdots, v_k\}$ since $0 = 0v_1 + 0v_2 + \cdots + 0v_k$.

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5.4.2 v_i is in the span

$v_i \in \text{span}\{v_1, v_2, \cdots, v_k\}$ since $v_i = 0v_1 + 0v_2 + \cdots + 1v_i + \cdots + 0v_k$.

5.4.3 Affine line

Given $v \in \mathbb{F}^n \backslash \{0\}$, the set $\text{span}\{v\} = \{tv | t \in \mathbb{F}\}$ is called a line through O.

In general, given v_0, and v in $\mathbb{F}^n$ with $v \neq 0$, $\{v_0 + tv | t \in \mathbb{F}\}$ is called a line through v_0 with direction v or affine line.

5.4.4 Affine plane

Given v, w in $\mathbb{F}^n \backslash \{0\}$ s.t. $v \neq \alpha w$ for any $\alpha \in \mathbb{F}$, the set $\text{span}\{v, w\} = \{av + bw | a, b \in \mathbb{F}\}$ is called a plane through O generated by v and w.

In general, given v_0, v and w in $\mathbb{F}^n$ with $v \neq \alpha w$ for any $\alpha \in \mathbb{F}$. The set $\{v_0 + av + bw | a, b \in \mathbb{F}\}$ is called affine plane.

5.4.5 Canonical vector

Consider the vector $v = \begin{bmatrix} a_1 \\ a_2 \\ \cdots \\ a_n \end{bmatrix}$ in $\mathbb{F}^n$. We can write v as a linear combination of the canonical vectors:

$$v = a_1 e_1 + a_2 e_2 + \cdots + a_n e_n \tag{21}$$

Where e_i is the vector whose i-th component is 1 and all other components are 0.

We call $e_1, e_2, \cdots, e_n$ the canonical vectors.

6 Matrices

6.1 Definition of matrices

An $m \times n$ matrix A with coefficients in $\mathbb{F}$ is a rectangular array witm m rows, n columns and mn elements in $\mathbb{F}$.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \tag{22}$$

The set of $m \times n$ matrices over $\mathbb{F}$ is denoted by $\mathbb{F}^{m\times n}$ or $M_{m\times n}(\mathbb{F})$.

If $m = n$, we say that $\mathbb{F}^{n\times n}$ is the set of square matrices with size n.

We can say that a matrix A in $\mathbb{F}^{m\times n}$ is a collection of n column vectors u_1, u_2, $\cdots$, u_n in $\mathbb{F}^m$. That we usually write:

$$A = [u_1|u_2|\cdots|u_n] \tag{23}$$

6.2 Matrices addition, scalar multiplication and matrix times a vector

6.2.1 Definition of addition

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices in $\mathbb{F}^{m\times n}$. The sum $A + B$ is the $m \times n$ matrix $C = [c_{ij}]$ whose i, j entry is:

$$\begin{bmatrix} c_11 & c_{12} & \cdots & c_{1n} \\ c_21 & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix} = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \cdots & a_{1n}+b_{1n} \\ a_{21}+b_{21} & a_{22}+b_{22} & \cdots & a_{2n}+b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}+b_{m1} & a_{m2}+b_{m2} & \cdots & a_{mn}+b_{mn} \end{bmatrix} \tag{24}$$

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6.2.2 Definition of scalar multiplication

Let $A = [a_{ij}]$ be a matrix in $\mathbb{F}^{m\times n}$ and let $c \in \mathbb{F}$. The product cA is the $m \times n$ matrix $D = [d_{ij}]$ whose i, j entry is:

$$\begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1n} \\ d_{21} & d_{22} & \cdots & d_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ d_{m1} & d_{m2} & \cdots & d_{mn} \end{bmatrix} = \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \cdots & ca_{mn} \end{bmatrix} \tag{25}$$

6.2.3 Definition of matrix times a vector

Let $A = [a_{ij}]$ be a matrix in $\mathbb{F}^{m\times n}$ and let $v = \begin{bmatrix} v_1 \\ v_2 \\ \cdots \\ v_n \end{bmatrix}$ be a vector in $\mathbb{F}^n$. The product Av is the vector $u = \begin{bmatrix} u_1 \\ u_2 \\ \cdots \\ u_m \end{bmatrix}$ in $\mathbb{F}^m$ whose i-th entry is:

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} = \begin{bmatrix} a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \cdots + a_{mn}v_n \end{bmatrix} \tag{26}$$

$$Av = [Ae_1|Ae_2|\cdots|Ae_n] \begin{bmatrix} v_1 \\ v_2 \\ \cdots \\ v_n \end{bmatrix} = v_1(Ae_1) + v_2(Ae_2) + \cdots + v_n(Ae_n)$$

6.3 Properties of addition, scalar multiplication and matrix times a vector

6.3.1 Properties of addition

Let A, B and C be three matrices in $\mathbb{F}^{m\times n}$. Then

$$A + B = B + A \tag{27}$$

$$(A + B) + C = A + (B + C) \tag{28}$$

There exists a unique matrix O in $\mathbb{F}^{m\times n}$ s.t.:

$$A + O = A \tag{29}$$

For each matrix A in $\mathbb{F}^{m\times n}$, there exists a unique matrix $-A$ in $\mathbb{F}^{m\times n}$ s.t.:

$$A + (-A) = O \tag{30}$$

6.3.2 Properties of scalar multiplication

Let A, B be two matrices in $\mathbb{F}^{m\times n}$ and let c, d be two elements in $\mathbb{F}$. Then

$$c(dA) = (cd)A \tag{31}$$

$$1A = A \tag{32}$$

$$c(A + B) = cA + cB \tag{33}$$

$$(c + d)A = cA + dA \tag{34}$$

6.3.3 Properties of matrix times a vector

Let A be a matrix in $\mathbb{F}^{m\times n}$, let v, w be two vectors in $\mathbb{F}^n$ and let c be an element in $\mathbb{F}$. Then

$$A(v + w) = Av + Aw \tag{35}$$

$$A(cv) = c(Av) \tag{36}$$

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6.4 Matrices multiplication

6.4.1 Definition

Let $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times p}$. The product AB is the $m \times p$ matrix C whose i, j entry is

$$c_{ij} = \sum_{r=1}^{n} = a_{ic}b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \tag{37}$$

$AB = [A(Be_1) \mid A(Be_2) \mid \cdots \mid A(Be_p)]$

6.4.2 Example

$A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & -1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 & -1 & 1 \\ -1 & 0 & 1 & -1 \\ 1 & 0 & -1 & 1 \end{bmatrix}$

$AB = \begin{bmatrix} 1 & 1 & -2 & 2 \\ 0 & -2 & 2 & -2 \end{bmatrix}$

6.4.3 Remark

Let A be a $m \times n$ matrix on $\mathbb{F}$, then

- $I_{m\times m}A = A$
- $AI_{n\times n} = A$
- $O_{m\times n}A = O_{m\times p}$
- $AO_{n\times p} = O_{m\times p}$

6.4.4 Properties

Those properties of matrix multiplication can also shift to linear transformations.

Associative Law

Let three matrices A, B and C such that the products are defined. Then

$$A(BC) = (AB)C = ABC \tag{38}$$

Distributive Law

Let A, B and C be three matrices such that the products are defined. Then

$$A(B + C) = AB + AC \tag{39}$$

$$(A + B)C = AC + BC \tag{40}$$

Power of a Matrix

Let $A \in \mathbb{F}^{n\times n}$. Define $A^0 = I_n$ and for $k \geq 1$, define $A^k = AA^{k-1}$.

Not Commutative

The matrix multiplication is not commutative in general. i.e., $AB \neq BA$ in general.

Multiplication with Transpose

Let $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times p}$. Then

$$(AB)^T = B^T A^T \tag{41}$$

6.5 Elementary row operations

6.5.1 Types of elementary row operations

We have three types of elementary row operations:

- Type 1: Multiply a row by a nonzero scalar.
- Type 2: Add to one row a scalar multiple of another row.
- Type 3: Interchange two rows.

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Two matrices are row-equivalent if one can be obtained from the other by a sequence of elementary row operations.

6.5.2 Theorem of existence of inverse operation

The inverse operation (function) of an elementary row operation exists and is an elementary row operation of the same type.

6.6 Row-echelon and row-reduced echelon forms

6.6.1 Definition of row-echelon form

A matrix is in row-echelon form if:

- All nonzero rows are above any rows of all zeros.
- The leading coefficient of a nonzero row is always strictly to the right of the leading coefficient of the previous row.

6.6.2 Definition of row-reduced echelon form

A matrix is in row-reduced echelon form if:

- It is in row-echelon form.
- The leading coefficient in each nonzero row is 1.
- Each leading 1 is the only nonzero entry in its column.

6.6.3 Remark

Every matrix is row-equivalent to one and only one row-reduced echelon matrix.

6.7 System of linear equations

6.7.1 Definition of argumented matrix

Given a system of linear equations, its argumented matrix is the matrix obtained by writing the coefficients of the variables and the constants in a rectangular array.

$$Ax = b : \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] \tag{42}$$

We call it homogeneous if $b = 0$.

6.7.2 Remark

Two systems of linear equations are equivalent if each equation in each system is a linear combination of the equations in the other system.

6.7.3 Definition of augmented matrix

Let $Ax = b$ be a linear system with m equations and n unknowns. Let $M = [A|b]$ be its augmented matrix.

6.8 Equivalent

6.8.1 Definition of equivalent

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For systems of linear equations, two systems of linear equations are equivalent if each equation in one system is a linear combination of the equations in the other system, and vice versa.

If A and B are $m \times n$ matrices, we sat that B is row equivalent to A if B can be obtained from A by a sequence of elementary row operations. In this case, we also say that A is equivalent to B.

We also have a theorem that, equivalent systems of linear equations have exactly the same solutions.

6.8.2 Remark

The equivalence of matrices is an equivalence relation. They have the following properties:

- Reflexive: A is equivalent to A.
- Symmetric: If A is equivalent to B, then B is equivalent to A.
- Transitive: If A is equivalent to B and B is equivalent to C, then A is equivalent to C.

6.9 Consistency of a linear system

6.9.1 Definition of consistent

Let $Ax = b$ be a l.s. and let $[R|b']$ be the row-reduced echelon form from $[A|b]$. If there's a zero row r_i in R s.t. $b'_i \neq 0$. Thus $Ax = b$ has no solution, and it's inconsistent.

Otherwise, $Ax = b$ has at least one solution, and it's consistent.

6.10 Determined

If the system is consistent and if every column of R is a pivot column. We say that the system is determined, and it has a unique solution.

6.11 independent variables

If the system is consistent and there exists a column without leading 1's, then the variable given by that column is called an independent variable and the variables given by the pivots columns are called dependent variables.

6.12 Invertible matrices

6.12.1 Definition of invertible matrices

Let $A \in \mathbb{F}^{n\times n}$. An $n \times n$ matrix B s.t. $BA = I$ is called a left-inverse of A. An $n \times n$ matrix B s.t. $AB = I$ is called a right-inverse.

If $AB = BA = I$, then B is called a two-sided inverse of A and A is said to be invertible.

6.12.2 Lemma: invertible matrices

If A has a left-inverse B and a right-inverse C, then $B = C$.

The proof is in the "Proofs" section.

6.12.3 Notation of inverse matrix

If A is invertible, A^{-1} denotes its inverse.

6.12.4 Theorem inverse of inverse matrix is itself

Let $A \in \mathbb{F}^{n\times n}$. If A is invertible, then $(A^{-1})^{-1} = A$

6.12.5 Proposition: a invertible matrix is an isomorphism

A square matrix $A \in \mathbb{F}^{n\times n}$ is invertible iff $T_A : \mathbb{F}^n \to \mathbb{F}^n$ is an isomorphism.

Given A in $\mathbb{F}^{n\times n}$, T_A is an isomorphism iff rank A = n.

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6.12.6 Theorem: invertiblility and the rank

A square matrix A in $\mathbb{F}^{n\times n}$ is invertible iff rank$A = n$.

6.12.7 Remark: find the inverse of a matrix

A is invertible, then there exists unique B s.t. $AB = BA = I$.

To find A^{-1}, we can write $[A \mid I]$ and try to write as $[I \mid B]$ by row operations. Then $B = A^{-1}$.

6.13 Diagonal matrix

6.13.1 Definition of diagonal matrix

A matrix $A \in \mathbb{F}^{n\times n}$ is called a diagonal matrix if $a_{ij} = 0$ for all $i \neq j$.

6.13.2 Notation

A diagonal matrix A is usually denoted by $\mathrm{diag}(d_1, d_2, \cdots, d_n)$ where $d_i = a_{ii}$ for $i = 1, 2, \cdots, n$.

6.13.3 Proposition: invertibility of diagonal matrix

A diagonal matrix $\mathrm{diag}(d_1, d_2, \cdots, d_n)$ is invertible iff $d_i \neq 0$ for all $i = 1, 2, \cdots, n$. In this case,

$$\mathrm{diag}(d_1, d_2, \cdots, d_n)^{-1} = \mathrm{diag}\left(\frac{1}{d_1}, \frac{1}{d_2}, \cdots, \frac{1}{d_n}\right) \tag{43}$$

7 Vector space and subsapce

7.1 Definition of vector space

A **vector space** V over a field $\mathbb{F}$ consists of the following properties:

- A field $\mathbb{F}$ of scalars.
- A set V of objects, called vectors.
- An operation, called vector addition ($+ : V \times V \to V$)

$$\begin{aligned}
A_1 &: (u+v)+w = u+(v+w), \forall u,v,w \in V \\
A_2 &: u+v = v+u, \forall u,v \in V \\
A_3 &: \exists 0 \in V \text{ s.t. } 0+u = u, \forall u \in V \\
A_4 &: \forall u \in V, \exists -u \in V \text{ s.t. } u+(-u) = 0
\end{aligned}$$

- An operation, called scalar multiplication ($\cdot : \mathbb{F} \times V \to V$)

$$\begin{aligned}
P_1 &: 1 \cdot v = v, \forall v \in V \\
P_2 &: \alpha(\beta v) = (\alpha\beta)v, \forall \alpha,\beta \in \mathbb{F}, \forall v \in V \\
P_3 &: \alpha(u+v) = \alpha u + \alpha v, \forall \alpha \in \mathbb{F}, \forall u,v \in V \\
P_4 &: (\alpha+\beta)v = \alpha v + \beta v, \forall \alpha,\beta \in \mathbb{F}, \forall v \in V
\end{aligned}$$

By the way, to prove that a set V is a vector space, it's better to prove the addition and scalar multiplication are closed.

7.2 Examples of vector spaces

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7.2.1 $\mathbb{F}^n$ over $\mathbb{F}$

$\mathbb{F}^n$ over $\mathbb{F}$

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1+y_1 \\ x_2+y_2 \\ \vdots \\ x_n+y_n \end{pmatrix}, \quad \alpha \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{pmatrix}$$

7.2.2 $V = \{0\}$ over $\mathbb{F}$

$V = \{0\}$ over $\mathbb{F}$

$$0+0=0, \quad \alpha \cdot 0 = 0, \forall \alpha \in \mathbb{F}$$

7.2.3 $V = \mathbb{F}[x]$ over $\mathbb{F}$

$V = \mathbb{F}[x]$ over $\mathbb{F}$

$$\mathbb{F}[x] = \{p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \mid a_1, a_2, \ldots, a_n \in \mathbb{F}\}$$

$$p(x) = \sum_{i=0}^{n} a_i x^i, \quad q(x) = \sum_{i=0}^{m} b_i x^i$$

$$p(x) + q(x) = \sum_{i=0}^{\max\{n,m\}} (a_i + b_i) x^i$$

$$\alpha \cdot p(x) = \sum_{i=0}^{n} \alpha a_i x^i$$

7.2.4 $\mathbb{F}^{m\times n}$

The space of $m \times n$ matrices over the field $\mathbb{F}$, denoted $\mathbb{F}^{m\times n}$ (or $M_{m\times n}(\mathbb{F})$).

$$A, B \in \mathbb{F}^{m\times n}, \quad (A+B)_{ij} = a_{ij} + b_{ij}$$

$$\alpha \in \mathbb{F}, \quad (\alpha A)_{ij} = \alpha a_{ij}$$

7.2.5 function f

The space of functions from a set S to a field $\mathbb{F}$. $V = \{f \mid f : S \to \mathbb{F}, \text{f is a function}\}$ over $\mathbb{F}$

$$+ : V \times V \to V, \quad f + g : S \to \mathbb{F}, \quad (f+g)(x) = f(x) + g(x)$$

$$\cdot : \mathbb{F} \times V \to V, \quad \alpha \cdot f : S \to \mathbb{F}, \quad (\alpha \cdot f)(x) = \alpha f(x)$$

7.3 Definition of subspace

Let V be a vector space over a field $\mathbb{F}$. A subspace of V is a non-empty subset W of V which is itself a vector space over $\mathbb{F}$ with the operations of vector addition and scalar multiplication defined on V. We denote it as $W \leq V$.

7.4 Example of subspace

$$V = \mathbb{R}^3, \quad W = \{(x, y, 0) \mid x, y \in \mathbb{R}\}, \quad W \leq V$$

7.5 Proposition: test for subspace

A non-empty subset W of V is a subspace of V if and only if for each pair of vectors u, v in W and each scalar α in $\mathbb{F}$ satisfies:

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i : $0 \in W$

ii : $u + v \in W$

iii : $\alpha v \in W$

7.6 Examples of the test for subspace

7.6.1 The solution set of a homogeneous linear system

Let W be the solution set of a linear system $Ax = 0_m$ with $A \in M_{m\times n}(\mathbb{F})$. Then W is a subspace of $\mathbb{F}^n$.

i : $0_n \in W : A0_n = 0_m$

ii : $v, w \in W,$ then $A(v+w) = Av + Aw = 0_m + 0_m = 0_m,$ so $v + w \in W$

iii : $\alpha \in \mathbb{F}, v \in W :$ then $A(\alpha v) = \alpha(Av) = \alpha(0_m) = 0_m,$ so $\alpha v \in W$

Thus W is a subspace of $\mathbb{F}^n$ ($W \leq \mathbb{F}^n$).

7.6.2 $A \in \mathbb{F}^{n\times n}$ is symmetric

Let $V = \mathbb{F}^{n\times n}$, $W = \{A \in \mathbb{F}^{n\times n} \mid a_{ij} = a_{ji}, \forall 1 \leq i, j \leq n\}$.
$W \leq V$:

i : $0_{n\times n} \in W$

ii : $A, B \in W,$ then $(A+B)_{ij} = a_{ij} + b_{ij} = a_{ji} + b_{ji} = (A+B)_{ji}, \forall 1 \leq i, j \leq n,$ so $A + B \in W$

iii : $\alpha \in \mathbb{F}, A \in W :$ then $(\alpha A)_{ij} = \alpha a_{ij} = \alpha a_{ji} = (\alpha A)_{ji}, \forall 1 \leq i, j \leq n,$ so $\alpha A \in W$

Thus W is a subspace of V ($W \leq V$).

8 Span and generating system

8.1 Definition of span

A vector v in V is said to be a linear combination of the vectors $v_1, v_2, \ldots, v_n$ in V if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ in $\mathbb{F}$ such that

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = \sum_{i=1}^{n} \alpha_i v_i. \tag{44}$$

The set of all linear combinations of $v_1, v_2, \ldots, v_n$ is denoted by $\operatorname{span}\{v_1, v_2, \ldots, v_n\}$ and it's called the subspace spanned by the vectors $v_1, v_2, \ldots, v_n$. The set $\{v_1, v_2, \ldots, v_n\}$ is called a **generating system** of the subspace $\operatorname{span}\{v_1, v_2, \ldots, v_n\}$.

Notice that $\operatorname{span}\{v_1, v_2, \ldots, v_n\}$ is a subspace of V:

i : $0 \in \operatorname{span}\{v_1, v_2, \ldots, v_n\}$, since $0 = 0v_1 + 0v_2 + \cdots + 0v_n$

ii : $v, w \in \operatorname{span}\{v_1, v_2, \ldots, v_n\}$, then $v + w \in \operatorname{span}\{v_1, v_2, \ldots, v_n\}$

iii : if $v \in \operatorname{span}\{v_1, v_2, \ldots, v_n\}$, then $\alpha v \in \operatorname{span}\{v_1, v_2, \ldots, v_n\}$

Thus, $\operatorname{span}\{v_1, v_2, \ldots, v_n\}$ is a subspace of V.

In particular, if $V = \mathbb{F}^m$, then $\operatorname{span}\{v_1, v_2, \ldots, v_n\} = \{b \in \mathbb{F}^m \mid Ax = b\}$ where $A = [v_1 \mid v_2 \mid \cdots \mid v_n]$.

8.2 Example of span

8.2.1 $\mathbb{R}^2$

$\mathbb{F} = \mathbb{R}, V = \mathbb{R}^2, S = \{(1,0),(0,1)\}, \forall (x,y) \in \mathbb{R}^2,\ (x,y) = x(1,0)+y(0,1)$ then $\mathbb{R}^2 = \operatorname{span}\{(1,0),(0,1)\}$.

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8.2.2 $\mathbb{F}^n$

$V = \mathbb{F}^n, S = \{e_1, e_2, \cdots, e_n\}$ where e_i is the vector whose i-th component is 1 and all other components are 0. Then $\mathbb{F}^n = \operatorname{span}\{e_1, e_2, \cdots, e_n\}$.

8.2.3 $\mathbb{F}^{m\times n}$

In general, $M_{m\times n}(\mathbb{F}) = \operatorname{span}\{E^{ij} \mid 1 \le i \le m, 1 \le j \le n\}$ where E^{ij} is the matrix whose (i,j)-th entry is 1 and all other entries are 0.

8.2.4 $\mathbb{F}[x]$

V=$\mathbb{F}[x]$, then:

$$\mathbb{F}[x] = \operatorname{span}\{x^i \mid i \in \mathbb{N}_0\}$$

8.3 Extension: smallest subspace containing given vectors

$\operatorname{span}\{v_1, v_2, \cdots, v_n\}$ is the smallest subspace of V containing $v_1, v_2, \cdots, v_n$.

8.4 Proposition: intersection of subspaces

Let V be a vector space over a field $\mathbb{F}$ and let S and T be two subspaces, then $S \cap T$ is also a subspace of V.

Proof see section "Proofs".

8.5 Remark: union of subspaces is not a subspace

If V is a vector space and $S, T \le V$. In general $S \cup T$ is not a subspace of V.

8.6 Example of the remark

$V = \mathbb{R}^2$, $S = \{(x, 0) \mid x \in \mathbb{R}\}$, $T = \{(0, y) \mid y \in \mathbb{R}\}$
$S \cup T = \{(x, 0) \mid x \in \mathbb{R}\} \cup \{(0, y) \mid y \in \mathbb{R}\}$
$(1, 0) \in S \cup T$, $(0, 1) \in S \cup T$, but $(1, 0) + (0, 1) = (1, 1) \notin S \cup T$
So $S \cup T$ is not a subspace of V.

8.7 Proposition: rank and generating system

A collection of vectors $v_1, v_2, \cdots, v_n \in \mathbb{F}^m$ is a generating set of V iff the matrix A=$[v_1|v_2|\cdots|v_n]$ has rank m. In this case m $\leq$ n.
Proof see section "Proofs".

9 Linear dependence and independence

9.1 Definition of linear dependence

Let V be a vector space over a field $\mathbb{F}$. A subset S of V is said to be linearly dependent (or l.d.) if there exist vectors $v_1, v_2, \cdots, v_n \in S$ and scalars $\alpha_1, \alpha_2, \cdots, \alpha_n \in \mathbb{F}$, not all of which are zero, such that

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0. \tag{45}$$

A set which is not linearly dependent (l.d.) is called linearly independent (l.i.).
We sometimes say that the vectors are dependent (or independent).

9.2 Remark: linear dependence with finite set

For a set S, with n elements, S is l.i. if

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0 \text{ iff } \alpha_1 = \alpha_2 = \cdots = \alpha_n = 0. \tag{46}$$

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9.3 Example of linear dependence and independence

9.3.1 $V = \mathbb{R}^2$

$V = \mathbb{R}^2$ The vectors (1,2) and (-1,2) are l.i..

9.3.2 $\mathbb{R}^n$

In $\mathbb{R}^n$, the set $\{e_1, e_2, \cdots, e_n\}$ is l.i..

9.3.3 $\mathbb{F}_n[x]$

In $\mathbb{F}_n[x]$, the set $\{1, x, x^2, \cdots, x^n\}$ is l.i..

9.4 Another way to prove linear independent

We can write all the vectors as rows of a matrix, then use row operations to convert the matrix to its row echelon form. If we obtain a zeros row, then the vectors are linear dependent, otherwise they are linear independent.

9.5 Proposition: properties of linear dependence and independence

- Any set which contains the zero vector is l.d..
- Let S be a l.d. set. If T is a subset of V such that $S \subseteq T$, then T is l.d..
- If $S = \{v_1, v_2, \cdots, v_n\}$ is a l.d. set with $n \geq 2$, there exists $i \in \{1, 2, \ldots, n\}$ such that v_i is a linear combination of the other vectors.
- Let S be a l.i. set, if $S' \subseteq S$, then S' is l.i..
- A set S is l.i. iff each finite subset of S is l.i..
- $\mathbb{F}[x] = \text{span}\{1, x, x^2, \cdots, x^i, \cdots\} = \text{span}\{x^i \mid i \in \mathbb{N}_0\}$, S is l.i..

9.6 Examples of the proposition

9.6.1 $v \neq \alpha u$

Let $S = \{u, v\}$, S is l.i. iff $u \neq 0$ and $v \neq \alpha u \quad \forall \alpha \in \mathbb{F}$

9.6.2 $w \notin \operatorname{span}\{u, v\}$

Let $S = \{u, v, w\}$. S is l.i. iff $\{u, v\}$ is a l.i. set and $w \notin \operatorname{span}\{u, v\}$

9.7 Proposition: a criterion for linear independence

Let $v_1, v_2, \cdots, v_r$ be vectors such that

- $v_1 \neq 0$,
- $v_i \notin \operatorname{span}\{v_1, v_2, \cdots, v_{i-1}\}$ for all $i = 2, \ldots, r$.

Then $\{v_1, v_2, \cdots, v_r\}$ is a linearly independent set.

9.8 Proposition: rank and linear independence

A set of vectors $v_1, v_2, \cdots, v_n$ in $\mathbb{F}^m$ is l.i. if and only if the matrix $A = [v_1|v_2|\cdots|v_n]$ has rank n.
Proof see section "Proofs".

10 Basis

10.1 Definition of basis

Let V be a vector space over a field $\mathbb{F}$. A basis for V is a l.i. set B s.t.

$$\operatorname{span}\{B\} = V. \tag{47}$$

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10.2 Example of basis (V=$\mathbb{R}^2$)

10.2.1 $\{(1, 0), (0, 1), (1, 1)\}$

$B = \{(1, 0), (0, 1), (1, 1)\}$ is not a basis for $\mathbb{R}^2$. Because B is not l.i. .

10.2.2 $\{(1, 1), (1, -1)\}$

$S = \{(1, 1), (1, -1)\}$ is a l.i. set and $\operatorname{span} S = \mathbb{R}^2$, so S is a basis for $\mathbb{R}^2$.
(x,y) = $\frac{x+y}{2}$(1,1) + $\frac{x-y}{2}$(1,-1)

10.3 Extension: unique representation of vectors in terms of basis

A set of vectors $v_1, v_2, \cdots, v_n$ is a basis for the vector space V iff any $v \in V$ can be written uniquely as a l.c. $v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$.
That is for each $v \in V$,

$$\exists! \alpha_1, \alpha_2, \cdots, \alpha_n \in \mathbb{F} \text{ s.t. } v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n \tag{48}$$

10.4 Proposition: characterization of basis for $\mathbb{F}^m$

A set of n vectors $v_1, v_2, \cdots, v_n \in \mathbb{F}^m$ is a basis for $\mathbb{F}^m$ iff $n = m$ and the rank of $A = [v_1|v_2|\cdots|v_n]$ is n.
Proof see section "Proofs".

10.5 Theorem: characterization of basis in general

Let V be a f.d.v.s. over a field $\mathbb{F}$ with $dim_{\mathbb{F}}V = n$. Let $\{v_1, v_2, \cdots, v_n\}$ be a set of vectors in V. Then the following conditions are equivalent:

(1) : $\{v_1, v_2, \cdots, v_n\}$ is a basis for V.

(2) : V=span$\{v_1, v_2, \cdots, v_n\}$

(3) : $\{v_1, v_2, \cdots, v_n\}$ is a l.i. set.

The proof see section "Proofs".

10.6 Example of the characterization of basis

The vectors $(1, -1)$ and $(1, 1)$ form a basis for $\mathbb{R}^2$. $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$, rank$(A) = 2$.

Therefore, $\{(1, -1), (1, 1)\}$ is a basis for $\mathbb{R}^2$.

10.7 Extension: span of subsets of a spanning set

If $w_1, w_2, \cdots, w_k$ are vectors in span$\{v_1, v_2, \cdots, v_n\}$, then span$\{w_1, w_2, \cdots, w_k\} \subseteq$ span$\{v_1, v_2, \cdots, v_n\}$.

10.8 Extension: add a vector already in the span

Let $\{v_1, v_2, \cdots, v_r\}$ be a set of vectors such that $v_r \in$ span$\{v_1, v_2, \cdots, v_{r-1}\}$.
Then

$$\text{span}\{v_1, \ldots, v_{r-1}, v_r\} = \text{span}\{v_1, \ldots, v_{r-1}\},$$

i.e. adding v_r does not enlarge the span.

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10.9 Theorem: basis subset of a finite generating set

Let $\{v_1, v_2, \cdots, v_n\}$ be a finite set of non-zero vectors in a vector space V. Then $\{v_1, v_2, \cdots, v_n\}$ contains a basis for the subspace span$\{v_1, v_2, \cdots, v_n\}$.
Proof see section "Proofs".

10.10 Proposition: existence of basis

Let $v_1, v_2, \cdots, v_n$ be vectors in $\mathbb{F}^m$ and $A = [v_1|v_2|\cdots|v_n] \in M_{m\times n}(\mathbb{F})$.
Let R be a row-reduced echelon form that is row-equivalent to A.
Let $i_1 < i_2 < \cdots < i_r$ be the positions of pivot columns.
Then $\{v_{i_1}, v_{i_2}, \cdots, v_{i_r}\}$ is a basis for span$\{v_1, v_2, \cdots, v_n\}$.
Proof see section "Proofs".

10.11 Theorem: extend a linearly independent sets to a basis

Let V be a f.d.v.s. over $\mathbb{F}$. Then any l.i. set $\{v_1, v_2, \cdots, v_k\}$of V can be extended to a basis of V.
The proof see section "Proofs".

11 Dimension

11.1 Definition of dimension

Let V be a finitely generated vector space (f.d.v.s.) over a field $\mathbb{F}$, and let $B = \{v_1, v_2, \cdots, v_n\}$ be a basis for V. The number n is called the dimension of V.

$$dim_{\mathbb{F}}V = n \tag{49}$$

11.2 Fundamental lemma of linear algebra

Let $\{v_1, v_2, \cdots, v_n\}$ be a linearly independent set of vectors in a vector space V over a field $\mathbb{F}$. Then any subset of $\text{span}\{v_1, v_2, \cdots, v_n\}$ with more than n vectors is linearly dependent.

11.3 Theorem: elements of basis have same number

If $\{v_1, v_2, \cdots, v_m\}$ and $\{w_1, w_2, \cdots, w_n\}$ are basis for a vector space V over a field $\mathbb{F}$, then $n = m$.
Proof see section "Proofs".

11.4 Example of dimension

- $dim_{\mathbb{R}}\mathbb{R}^n = n$
- $dim_{\mathbb{F}}M_{m\times n}(\mathbb{F}) = mn$
- $dim_{\mathbb{R}}\mathbb{C}^1 = 2$
- $dim_{\mathbb{R}}\mathbb{C}^n = 2n$

11.5 Definition of column space and row space

$A \in M_{m\times n}(\mathbb{F})$, $A = [v_1|v_2|\cdots|v_n]$ or $A = \begin{bmatrix} w_1 \\ w_2 \\ \cdots \\ w_m \end{bmatrix}$ where $v_i \in \mathbb{F}^m$, $1 \le i \le n$ and $w_j \in \mathbb{F}^n$, $1 \le j \le m$.
Column-space $= \text{span}\{v_1, v_2, \cdots, v_n\} \subseteq \mathbb{F}^m$.
Row-space $= \text{span}\{w_1, w_2, \cdots, w_m\} \subseteq \mathbb{F}^n$.

11.6 Definition of rank

Let A be in $M_{m\times n}(\mathbb{F})$. We difine the rank of A as the dimension of the subspace of $\mathbb{F}^m$ generated by the columns (column space) of A. We denote this number by rk A or rank A.

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11.7 Remark: rank and non-zero rows/pivot columns

The definition of the rank is equivalent to the number of non-zero rows in the row-reduced echelon form R that is row-equivalent to A or the number of pivot columns in R.

11.8 Two properties

11.8.1 Row equivalent matrices have the same rank

If A and A' are row equivalent, then they have the same rank.

11.8.2 rank A = rank R

If R is a row-reduced echelon form that is row equivalent to A, then rank A = rank R.

11.9 Theorem: dimension of subspace is less than or equal to the dimension of the space

Let W be a subspace of a f.d.v.s. V over a field $\mathbb{F}$. Then W is also a f.d.v.s. and $dim_{\mathbb{F}}W \le dim_{\mathbb{F}}V$.
The proof see section "Proofs".

12 The sum of subsets, subspaces and direct sum

12.1 Definition of the sum of subsets

Let V be a vector space over a field $\mathbb{F}$ and let S and T be two subsets of V. The sum of S and T is the set:

$$S + T = \{s + t \mid s \in S, t \in T\} \tag{50}$$

12.2 Definition of the sum of subspaces

Let W_1 and W_2 be two subspaces in a vecotr space V. We define the sum $W_1 + W_2$ as

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1 \text{ and } w_2 \in W_2\} \tag{51}$$

Notice that $W_1 \subseteq W_1 + W_2$, $W_2 \subseteq W_1 + W_2$, and $W_1 \cup W_2 \subseteq W_1 + W_2$.
In general, if W_1, W_2, $\cdots$, W_k are subspaces of a vector space V over $\mathbb{F}$, then

$$\sum_{i=1}^{k} W_i = W_1 + W_2 + \cdots + W_k = \{w_1 + w_2 + \cdots + w_k \mid w_i \in W_i \text{ for } i = 1, 2, \cdots, k\} \tag{52}$$

12.3 Proposition: the sum of subspaces is a subspace

If $W_1, W_2, \cdots, W_k$ are subspaces of a vector space V over a field $\mathbb{F}$, then the sum $W_1 + W_2 + \cdots + W_k$ is a subspace of V over $\mathbb{F}$.
Proof see section "Proofs".

12.4 Remark: sum of subsapces is the span of their bases' union

$B_U = \{u_1, u_2, \cdots, u_n\}$, $B_W = \{w_1, w_2, \cdots, w_m\} = \{u_{n+1}, u_{n+2}, \cdots, u_{n+m}\}$. And $U = \text{span}\{B_U\}$, $W = \text{span}\{B_W\}$. Then, a general element in $U + W$ is of the form:

$$\sum_{i=1}^{n+m} \gamma_i u_i = \sum_{i=1}^{n} \alpha_i u_i + \sum_{j=1}^{m} \beta_j w_j \tag{53}$$

$\Rightarrow U + W = \text{span}\{B_U \cup B_W\}$.

12.5 Theorem: the dimension of the sum of subspaces

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If W_1 and W_2 are subsapces of a f.d.v.s. V over $\mathbb{F}$, then

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2) \tag{54}$$

Proof see [104016 L-251222]-P14.
Also, we can know that $\dim_{\mathbb{F}}(\sum_{i=1}^{k} W_i) \leq \sum_{i=1}^{k} \dim_{\mathbb{F}}(W_i)$

12.6 Definition of direct sum

Let W_1, W_2, $\cdots$, W_k be subspaces of a vector space V over a field $\mathbb{F}$. Then $\sum_{i=1}^{k} W_i$ is called a direct sum if the following condition holds.

$$w_1 + w_2 + \cdots + w_k = 0 \text{ with } w_i \in W_i \Rightarrow w_i = 0, \forall 1 \leq i \leq k \tag{55}$$

In this case, we write $W_1 \oplus W_2 \oplus \cdots \oplus W_k$

12.7 Proposition: equivalent statements for direct sum

Let W_1, W_2, $\cdots$, W_k be subspaces of a vector space V over a field $\mathbb{F}$. Then the following statements are equivalent.

(i) $W_1 + W_2 + \cdots + W_k$ is a direct sum ($\bigoplus_{i=1}^{k} W_i$).

(ii) If $\mathcal{B}_i$ is a basis for W_i, then $\bigcup_{i=1}^{k} \mathcal{B}_i$ is a basis for $\sum_{i=1}^{k} W_i$.

(iii) $\dim_{\mathbb{F}}(\sum_{i=1}^{k} W_i) = \sum_{i=1}^{k} \dim_{\mathbb{F}}(W_i)$

Proof see [104016 L-251222]-P7.

12.8 Proposition: a criterion for direct sum of two subspaces

If W_1, W_2 are subspaces of a vector space V over a field $\mathbb{F}$, then $W_1 \oplus W_2$ if and only if $W_1 \cap W_2 = \{0\}$. Proof see [104016 L-251222]-P10.

12.9 Definition of complement

Let W be a subsapce of a v.s. V over $\mathbb{F}$. A subspace W' is called a **complement** of W in V if $V = W \oplus W'$. That is, $V = W' + W$ and $W' \cap W = \{0\}$.

12.10 Remark: dimension of the complement

If V is a f.d.v.s. over $\mathbb{F}$ and W' is the complement of W in V, then

$$\dim W' = \dim V - \dim W \tag{56}$$

13 Null space and kernel

13.1 Definition of null space or kernel

Let A be in $M_{m\times n}(\mathbb{F})$. The null space or kernel of A is the set of vectors v in $\mathbb{F}^n$ such that $Av = 0_m$. We denote it by $\ker(A)$.

$$\mathrm{Ker}(A) = \{v \in \mathbb{F}^n \mid Av = 0_m\} \tag{57}$$

$\mathrm{Ker}(A)$ = solution set of $Ax = 0$

13.2 Examples of kernel

13.2.1 Example of a square matrix

$$A = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

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$(x_1, x_2, x_3, x_4) \in$ Ker A iff A $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$$\begin{cases} x_1 = x_4 \\ x_2 = -3x_4 \\ x_3 = -2x_4 \end{cases}$$

Therefore $(x_1, x_2, x_3, x_4) = x_4$(1,-3,-2,1), Ker $(A) = \mathrm{span}\{(1, -3, -2, 1)\}$

13.2.2 Example of a non-square matrix

$$A = \begin{bmatrix} 1 & 3 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$(x_1, x_2, x_3, x_4) \in$ Ker A iff A $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\begin{cases} x_1 = -3x_2 - x_4 \\ x_3 = -2x_4 \end{cases}$$

Therefore (x_1, x_2, x_3, x_4)=x_2(-3,1,0,0)+x_4(-1,0,-2,1), $\mathrm{Ker}(A) = \mathrm{span}\{(-3, 1, 0, 0), (-1, 0, -2, 1)\}$

13.3 Rank-Nullity Theorem

Let A be in $M_{m\times n}(\mathbb{F})$. Then

$$\mathrm{rank}(A) + \dim_{\mathbb{F}}(\mathrm{Ker}(A)) = n \tag{58}$$

The proof see section "Proofs".

14 Linear transformation

14.1 Definition of linear transformation

Let V and W be vector spaces over a field $\mathbb{F}$. A linear transformation(l.t.) from V into W is a function T s.t.

$$T(u+v) = T(u) + T(v) \quad \forall u, v \in V \tag{59}$$

$$T(\alpha v) = \alpha T(v) \quad \forall \alpha \in \mathbb{F}, v \in V \tag{60}$$

$$T(\alpha u + v) = \alpha T(u) + T(v) \quad \forall \alpha, \in \mathbb{F}, u, v \in V \tag{61}$$

14.2 Remark: linear transformation of zero is always zero

If $T : V \to W$ is a l.t., then $T(0_V) = 0_W$.
$0_w = T(0_v) = T(0_v + 0_v) = T(0_v) + T(0_v) \Rightarrow T(0_v) = 0_W$.

14.3 Remark: linear transformation represented by matrix

If V is a l.c. of vectors $v_1, v_2, \cdots, v_n$ in V.
Then v=$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$, and if T is a l.t. from V into W, then

$$T(v) = \alpha_1 T(v_1) + \alpha_2 T(v_2) + \cdots + \alpha_n T(v_n) \tag{62}$$

In particular, if V=$\mathbb{F}^n$, $W = \mathbb{F}^m$ and T(e_1)=$\begin{bmatrix} a_{11} \\ a_{21} \\ \cdots \\ a_{m1} \end{bmatrix}$, T($e_2$)=$\begin{bmatrix} a_{12} \\ a_{22} \\ \cdots \\ a_{m2} \end{bmatrix}$, $\cdots$, T(e_n)=$\begin{bmatrix} a_{1n} \\ a_{2n} \\ \cdots \\ a_{mn} \end{bmatrix}$

Then, for any vector V in $\mathbb{F}^n$

$$v = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n \tag{63}$$

We have

$$T(v) = x_1 T(e_1) + x_2 T(e_2) + \cdots + x_n T(e_n) \tag{64}$$

$$= x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \cdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \cdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \cdots \\ a_{mn} \end{bmatrix} \tag{65}$$

$$= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \tag{66}$$

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14.4 Proposition: linear transformation relative to standard bases

If $T : \mathbb{F}^n \to \mathbb{F}^m$ is a l.t., then there exists a matrix A in $M_{m\times n}(\mathbb{F})$ s.t. $T(x) = Ax$ with $A = [T(e_1)|T(e_2)|\cdots|T(e_n)]$.
The matrix A is called the matrix of T relative to the standard bases.

14.5 Example: linear transformation represented by matrix

A counterclockwise rotation by an angle θ on $\mathbb{R}^2$ is a l.t. T:$\mathbb{R}^2 \to \mathbb{R}^2$
T(e_1)= $\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$, T($e_2$)= $\begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$, then T(x,y)= $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

14.6 Theorem of unique transformation

Let V be a f.d.v.s. over $\mathbb{F}$ and let $\{v_1, v_2, \cdots, v_n\}$ be an ordered basis for V. Let W be an v.s. over $\mathbb{F}$ and let $w_1, w_2, \cdots, w_n$ be any vecotrs in W. Then there is a unique linear transformation T from V into W, s.t. T(v_i)=w_i $\forall i = 1, 2, \cdots, n$.

14.7 Theorem of the algebra of linear transformations

Let V and W be two $\mathbb{F}-v.s.$ and let T_1 and T_2 be l.t. from v into W. The function $(T_1+T_2)(v)$=T_1(v)+T_2(v) and $(\alpha T_1)(v)$=αT_1(v) are l.t. from V into W.
The proof see section "Proofs".

14.8 Example of the algebra of linear transformations

Let A=$[v_1 \mid v_2 \mid \cdots \mid v_n]$, B=$[w_1 \mid w_2 \mid \cdots \mid w_n]$ be in $\mathbb{F}^{m\times n}$.
Then A+B=$[v_1+w_1 \mid v_2+w_2 \mid \cdots \mid v_n+w_n]$.
Consider $T_A:\mathbb{F}^n\to\mathbb{F}^m$ and $T_B:\mathbb{F}^n\to\mathbb{F}^m$ defined by $T_A(x)=Ax$ and $T_B(x)=Bx$ for $x\in\mathbb{F}^n$.
Then $T_A+T_B:\mathbb{F}^n\to\mathbb{F}^m$ is defined by $x\mapsto(T_A+T_B)(x)$ for $x\in\mathbb{F}^n$.

$$\begin{aligned}
(T_A+T_B)(x) &= T_A(x)+T_B(x)\\
&= Ax+Bx\\
&= [v_1 \mid v_2 \mid \cdots \mid v_n]x+[w_1 \mid w_2 \mid \cdots \mid w_n]x\\
&= (x_1v_1+x_2v_2+\cdots+x_nv_n)+(x_1w_1+x_2w_2+\cdots+x_nw_n)\\
&= x_1(v_1+w_1)+x_2(v_2+w_2)+\cdots+x_n(v_n+w_n)\\
&= [v_1+w_1 \mid v_2+w_2 \mid \cdots \mid v_n+w_n]x\\
&= (A+B)x\\
&= T_{A+B}(x)
\end{aligned}$$

$\alpha\in\mathbb{F}$

$$\begin{aligned}
(\alpha T_A)(x) &= \alpha T_A(x)=\alpha Ax=\alpha[v_1 \mid v_2 \mid \cdots \mid v_n]x\\
&= \alpha(x_1v_1+x_2v_2+\cdots+x_nv_n)\\
&= x_1(\alpha v_1)+x_2(\alpha v_2)+\cdots+x_n(\alpha v_n)\\
&= [\alpha v_1 \mid \alpha v_2 \mid \cdots \mid \alpha v_n]x\\
&= (\alpha A)x\\
&= T_{\alpha A}(x)
\end{aligned}$$

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14.9 Remark: zero linear transformation

The zero vector is the zero l.t. from V into W. O(v)=0_w

14.10 Proposition: composition of linear transformations

Let V,W and Z be three $\mathbb{F}-v.s.$. Let $T_1:V\to W$ and $T_2:W\to Z$ be two l.t..
Then the composition $T_2\circ T_1:V\to Z$ is a l.t.. $(T_2\circ T_1)(v)=T_2(T_1(v))$ for $v\in V$.
Proof see section "Proofs".

14.11 Remark: powers of linear transformation

$T:V\to V$, $T^2=T\circ T$, $T^3=T\circ T\circ T$, etc.
$T^0=id.$ for $T\neq O$.

14.12 Homomorphism

14.12.1 Definition of homomorphism

Let V and W be two $\mathbb{F}-v.s.$. The set of all l.t. from V into W is denoted by

$$\mathcal{L}(V,W)=Hom_{\mathbb{F}}(V,W)=\{T:V\to W \mid T \text{ is a l.t.}\} \tag{67}$$

14.12.2 Proposition: homomorphism is a vector space

$(Hom_{\mathbb{F}}(V,W),+,\cdot)$ is an $\mathbb{F}-v.s.$.

14.13 Monomorphism, epimorphism, isomorphism and endomorphism

14.13.1 Definition of monomorphism, epimorphism, isomorphism and endomorphism

Let V and W be two v.s. over $\mathbb{F}$ and let T be a l.t. from V into W.

i T is said to be a **monomorphism** if T is injective.

ii T is said to be an **epimorphism** if T is onto.

iii T is said to be an **isomorphism** if T is bijective.

iv T is said to be an **endomorphism** (of a linear operator) if $W = V$ $(T : V \to V)$.

We say that V and W are isomorphic (or W is isomorphic to V) if there exists an isomorphism from V into W. We denote $T \in \mathrm{End}(V)$ if $T : V \to V$ is an endomorphism.

14.13.2 Examples of monomorphism, epimorphism and isomorphism

i $T : \mathbb{F}^m \to \mathbb{F}^{m+1}$ defined by $x \mapsto T(x) = (x, 0)$. T is monomorphic.

ii $T : \mathbb{F}^m \to \mathbb{F}^{m-1}$ defined by $x \mapsto T(x) = (x_1, x_2, \cdots, x_{m-1})$. T is epimorphic.

iii $T : \mathbb{F}_n[x] \to \mathbb{F}^{n+1}$ defined by $p(x) \mapsto T(p(x)) = (a_0, a_1, a_2, \cdots, a_n)$. T is isomorphic.

14.13.3 Proposition: basis transformed by isomorphism

Let $T : V \to W$ be an isomorphism of $\mathbb{F} - v.s.$ and let $v_1, v_2, \cdots, v_n$ be vectors in V.

i $v_1, v_2, \cdots, v_n$ are l.i. iff $T(v_1), T(v_2), \cdots, T(v_n)$ are l.i..

ii $\mathrm{Span}\{v_1, v_2, \cdots, v_n\}$=$V$ iff $\mathrm{Span}\{T(v_1), T(v_2), \cdots, T(v_n)\}$=$W$.

iii $\{v_1, v_2, \cdots, v_n\}$ is a basis of V iff $\{T(v_1), T(v_2), \cdots, T(v_n)\}$ is a basis of W.

The proof see section "Proofs".

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14.13.4 Proposition: dimension of ispmorphic spaces

If V and W are isomorphic ($\cong$) finite dimensional v.s. over $\mathbb{F}$, then $\dim_{\mathbb{F}}(V) = \dim_{\mathbb{F}}(W)$.
The proof see section "Proofs".

14.13.5 Corollary: of the proposition dimension of isomorphic spaces

If $m \neq n$, then $\mathbb{F}^m \ncong \mathbb{F}^n$.

14.14 Theorem of isomorphism

14.14.1 Statement of the theorem

Let V be a vector space of dimension n over $\mathbb{F}$. Then $V \cong \mathbb{F}^n$.

14.14.2 Example of the theorem of isomorphism

i $\mathbb{R}_n[x] \cong \mathbb{F}^{n+1}$

ii $\mathbb{F}^{n\times m} \cong \mathbb{F}^{nm}$

14.14.3 Proposition: properties of isomorphisms

Let U, V, W be vector space over $\mathbb{F}$.

i The identity function id:$V \to V$ is a isomorphism.

ii If $T : V \to W$ is an isomorphism, then its inverse $T^{-1} : W \to V$ is also an isomorphism.

iii If $T_1 : U \to V$ and $T_2 : V \to W$ are isomorphisms, then the composition $T_2 \circ T_1 : U \to W$ is also an isomorphism.

14.15 Theorem: same dimension spaces are isomorphic

14.15.1 Statement of the theorem

Two vector spaces with same dimension over the field $\mathbb{F}$ are isomorphic.
The proof is in the "Proofs" section.

14.15.2 Remark: construction of isomorphism between same dimension spaces

If $B = \{v_1, v_2, \cdots, v_n\}$ is a basis for V and $B' = \{w_1, w_2, \cdots, w_n\}$ is a basis for W. Define $T : V \to W$ by $T(v_i) = w_i$, $\forall i \in \{1, 2, \cdots, n\}$, and T is an isomorphism.

14.16 Kernel of the linear transformation

14.16.1 Definition of the kernel of the linear transformation

Let V,W be two $\mathbb{F}$-v.s.. Let $T : V \to W$ be a l.t..

- The null space (or kernel) of T is the set of all vectors v in V s.t. $T(v) = 0_W$.

$$(\operatorname{null} T)\ \operatorname{Ker} T = \{v \in V \mid T(v) = 0\} \tag{68}$$

- The nullity is the dimension of the null space of T.
- The range (or Image) of T is the set of vectors w in W s.t. $T(v) = w$ for some v in V.

$$(\operatorname{range} T)\ \operatorname{Im} T = \{w \in W \mid T(v) = w \text{ for some } v \in V\} \tag{69}$$

- The rank of T is the dimension of the range of T.

$$(\operatorname{rank} T) = \dim(\operatorname{Im} T) \quad (= \dim(\operatorname{range} T)) \tag{70}$$

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14.16.2 Example of the kernel

$T : \mathbb{R}^3 \to \mathbb{R}^2$ s.t. $T(x_1, x_2, x_3) = (x_1, x_2)$.
$\operatorname{Ker} T = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid T(x_1, x_2, x_3) = (0, 0)\} = \{(0, 0, x_3) \mid x_3 \in \mathbb{R}\} = \operatorname{span}\{e_3\}$
$\dim(\operatorname{Ker} T) = \dim(\operatorname{span}\{e_3\}) = 1$
$\cdot\ \operatorname{Im} T = \{(x, y) \in \mathbb{R}^2 \mid \exists (x_1, x_2, x_3) \in \mathbb{R}^3 \text{ s.t. } T(x_1, x_2, x_3) = (x, y)\} = \mathbb{R}^2$
$\operatorname{rank} T = \dim(\operatorname{Im} T) = 2$
Notice that $\dim(\operatorname{Ker} T) + \dim(\operatorname{Im} T) = 1 + 2 = 3 = \dim(\mathbb{R}^3)$

14.17 The rank-nullity theorem over linear transformations

14.17.1 Statement of the theorem

Let V and W be two f.d.v.s. over $\mathbb{F}$, and let $T : V \to W$ be a l.t.. Then:

$$\dim(V) = \dim(\operatorname{Ker} T) + \dim(\operatorname{Im} T) = \operatorname{null} T + \operatorname{rank} T \tag{71}$$

Proof see [104016 L-251201]-P15

14.17.2 Corollary of the rank-nullity theorem

Let V and W be two n-dimensional v.s. over $\mathbb{F}$, and let $T : V \to W$ be a l.t.. Then the following statements are equivalent:

i T is isomorphism (bijective).

ii T is monomorphism (injective).

iii T is epimorphism (onto).

Proof see [104016 L-251201]-P20

14.18 Remark: images and inverse images of functions

Let $f : A \to B$ be a function. If $A' \subseteq A$, then $f(A') = \{b \in B \mid f(a) = b \text{ for some } a \in A'\}$. (Image of A', $\mathrm{Im}(f\mid_{A'})$)

In particular, if $A = A'$, $f(A) = Im(f) = \{b \in B \mid f(a) = b \text{ for some } a \in A\}$

If $B' \subseteq B$, then $f^{-1}(B') = \{a \in A \mid f(a) = b \text{ for some } b \in B'\}$ This is the inverse image of B'.

When $B' = \{b\}$, we write $f^{-1}(b)$.

14.19 Proposition: images and inverse images of linear transformations

Let $T : V \to W$ be a l.t. Then:

i If S is a subsapce of V, then $T(S)$ is a subspace of W. Moreover, if $S = \mathrm{span}\{v_1, v_2, \cdots, v_n\}$, then $T(S) = \mathrm{span}\{T(v_1), T(v_2), \cdots, T(v_n)\}$

ii If U is a subspace of W, then $T^{-1}(U)$ is a subsapce of V.

Proof see [104016 L-251201]-P5.

15 Coordinate and matrix rep. of linear transformation

15.1 Coordinate

15.1.1 Coordinate in $\mathbb{R}^3$

$\mathbb{R}^3 : \varepsilon = \{e_1, e_2, e_3\}$, $(x, y, z) = xe_1 + ye_2 + ze_3$, $[(x, y, z)]_\varepsilon = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

15.1.2 Definition of coordinate

If V is f.d.v.s. over $\mathbb{F}$, an ordered basis B for V is a finite sequence of vectors which is l.i. and spans V.

Suppose the V is a f.d.v.s. over $\mathbb{F}$, and let $B = \{v_1, v_2, \cdots, v_n\}$ be an ordered basis for V. Then for each $v \in V$, there are scalars $\alpha_1, \alpha_2, \cdots, \alpha_n \in \mathbb{F}$ s.t.

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n \tag{72}$$

We define $[-]_B : V \to \mathbb{F}^n$, $v \mapsto [v]_B = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$ as the coordinate vector of v relative to the basis B.

We can prove that $[v]_B + [w]_B = [v + w]_B$ and $c[v]_B = [cv]_B$.

Proof see [104016 L-251203]-P3.

15.1.3 Definition of coordinate matrix

$[v]_B = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$ is the coordinate matrix of V relative to the ordered basis B.

15.1.4 Proposition: coordinate is l.t. and isomorphism

The function $[-]_B : V \to \mathbb{F}^n$ is a l.t.. Moreover, $[-]_B$ is an isomorphism.

Proof see [104016 L-251203]-P4.

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15.1.5 Example of coordinate matrix

- Let $V = \mathbb{R}^n$, $\varepsilon = \{e_1, e_2, \cdots, e_n\}$. $x \in \mathbb{R}^n, [x]_\varepsilon = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$.
- $V = \mathbb{R}_n[x]$, $\varepsilon = \{1, x, x^2, \cdots, x^n\}$. $p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$, $[p(x)]_\varepsilon = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$.
- $V = \mathbb{R}^3$, $B = \{(1,1,1), (1,1,0), (1,0,0)\}$ an ordered basis of $\mathbb{R}^3$.
- $(x, y, z) = (\alpha_1 + \alpha_2 + \alpha_3, \alpha_1 + \alpha_2, \alpha_1)$. Thus $[(x, y, z)]_B = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} z \\ y - z \\ x - y \end{bmatrix}$

15.2 Representation of l.t. by Matrix

15.2.1 Flow chart of the process

The chart of the process is shown below:

$$\begin{tikzcd} v \arrow[r, "{[-]_B}"] \arrow[d, "T"'] & {[v]_B} \arrow[d, "{[T]^B_{B'}}"] \\ T(v) \arrow[r, "{[-]_{B'}}"'] & {[T(v)]_{B'}} \end{tikzcd}$$

Notice that $[T(v)]_{B'} = [T]^B_{B'}[v]_B$

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The process of derivation see [104016 L-251203]-P7.
At last we have: $A = [[T(v_1)]_{B_2} \mid [T(v_2)]_{B_2} \mid \cdots \mid [T(v_n)]_{B_2}]]$.

15.2.2 Definition of matrix representation of l.t.

The matrix $A = [[T(v_1)]_{B_2} \mid [T(v_2)]_{B_2} \mid \cdots \mid [T(v_n)]_{B_2}]]$ is called the matrix of T relative to the pair of ordered basis B_1 and B_2.
Notation: We write A as $[T]^{B_1}_{B_2}$ or $[T]_{B_1 B_2}$

15.2.3 Example

Example see [104016 L-251203]-P12.

15.2.4 Theorem: each l.t. corresponds to a unique matrix

Let V be an n-dim v.s. over $\mathbb{F}$ and W be an m-dim v.s. over $\mathbb{F}$. Let B be an ordered basis for V and let B' be an ordered basis for W. For each l.t. $T : V \to W$, there is an $m \times n$ matrix A over which entries in $\mathbb{F}$ s.t.

$$[T(v)]_{B'} = A[v]_B \tag{73}$$

For every vector v in V. Furthermore, $T \to A$ is a one to one correspondence between the set of all l.t. from V to W and the set of matrices of size $m \times n$, which entries in $\mathbb{F}$.
Proof see [104016 L-251203]-P18.

15.2.5 Corollary: kernel and image over matrix representation

i $\operatorname{Ker} T = \{v \in V \mid [v]_B \in \operatorname{Ker}[T]_{B'}^B\}$
In particular, $\{u_1, u_2, \cdots, u_r\}$ is a basis of $\operatorname{Ker} T$ iff $\{[u_1]_B, [u_2]_B, \cdots, [u_r]_B\}$ is a basis of $\operatorname{Ker}[T]_{B'}^B$

ii $\operatorname{Im} T = \{w \in W \mid [w]_{B'} \in \operatorname{Im}[T]_{B'}^B\} = \{w \in W \mid [w]_{B'} \in \operatorname{Col}([T]_{B'}^B)\}$
In particular, $\{u_1, u_2, \cdots, u_s\}$ is a basis for $\operatorname{Im} T$ iff $\{[u_1]_{B'}, [u_2]_{B'}, \cdots, [u_s]_{B'}\}$ is a basis for $\operatorname{Col}([T]_{B'}^B)$

15.2.6 Remark: addition and scalar multiplication of l.t.

If $T, T' : V \to W$ and $\alpha \in \mathbb{F}$, then
$[\alpha T + T']_{B'}^B(v) = [(\alpha T + T')(v)]_{B'} = [\alpha T + T']_{B'}^B[v]_B = [\alpha T(v) + T'(v)]_{B'} = \alpha[T(v)]_{B'} + [T'(v)]_{B'} = \alpha[T]_{B'}^B[v]_B + [T']_{B'}^B[v]_B = (\alpha[T]_{B'}^B + [T']_{B'}^B)[v]_B$
$[\alpha T + T']_{B'}^B = (\alpha[T]_{B'}^B + [T']_{B'}^B)[v]_B \ \forall v \in V \Rightarrow [\alpha T + T']_{B'}^B = \alpha[T]_{B'}^B + [T']_{B'}^B$

15.2.7 Theorem: martix representation is isomorphism

Let V be an n-d.v.s. over $\mathbb{F}$ and let W be an m-d.v.s. over $\mathbb{F}$. For each pair of ordered basis B, B' for V and W, representation

$$\text{The function } [-]_{B'}^B = \mathcal{F}_{BB'} : \operatorname*{Hom}_{\mathbb{F}}(V, W) \to \mathbb{F}^{m\times n}, T \mapsto [T]_{B'}^B \tag{74}$$

is an isomorphism.

15.2.8 Corollary: the dimension of Hom(V, W)

If $\dim_{\mathbb{F}} V = n$, $\dim_{\mathbb{F}} W = m$, then $\dim_{\mathbb{F}}(\operatorname{Hom}_{\mathbb{F}}(V, W)) = mn$.

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15.3 The compostition of linear transformation and matrix representation

15.3.1 Theorem: composition corresponds to matrix multiplication

Let V, W and Z be f.d. vector space over $\mathbb{F}$, let T_1 be a l.t. from V into W and T_2 be a l.t. from W into Z. If $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ are ord. bases for V, W, Z representation.
Then $[T_2 \circ T_1] = [T_2]_{\mathcal{B}_3}^{\mathcal{B}_2}[T_1]_{\mathcal{B}_2}^{\mathcal{B}_1}$.
Proof see [104016 L-251208]-P3.

15.3.2 Corollary: invertible l.t. corresponds to invertible matrix

Let V and W be two f.d.v.s. over $\mathbb{F}$. Let $T : V \to W$ be an invertible l.t.. If $\mathcal{B}_1$ and $\mathcal{B}_2$ are ordered bases for V and W respectively. Then $[T^{-1}]_{\mathcal{B}_1}^{\mathcal{B}_2} = ([T]_{\mathcal{B}_2}^{\mathcal{B}_1})^{-1}$.
Proof see [104016 L-251208]-P4.

15.4 Change of Basis

15.4.1 Introduction of change of basis

Suppose that V is a n-dimensional v.s. and $\mathcal{B}_1 = \{v_1, v_2, \cdots, v_n\}$ and $\mathcal{B}_2 = \{w_1, w_2, \cdots, w_n\}$ are two ordered bases for V.
There are unique scalars P_{ij} s.t. $w_j = P_{1j}v_1 + P_{2j}v_2 + \cdots + P_{nj}v_n$ for each $j = 1, 2, \cdots, n$.
Thus, if $v \in V$ and $v = \alpha_1 w_1 + \alpha_2 w_2 + \cdots + \alpha_n w_n$, $(\alpha_i \in \mathbb{F})$

$$\begin{aligned}
\text{Then} v &= \alpha_1(\sum_{i=1}^n P_{i1}v_i) + \alpha_2(\sum_{i=1}^n P_{i2}v_i) + \cdots + \alpha_n(\sum_{i=1}^n P_{in}v_i) \\
&= \sum_{j=1}^n(\alpha_j \sum_{i=1}^n P_{ij}v_i) \\
&= \sum_{i=1}^n\sum_{j=1}^n \alpha_j P_{ij}v_i
\end{aligned}$$

Therefore $V = \sum_{i=1}^{n}(\sum_{j=1}^{n}(P_{ij}\alpha_j))v_i = \sum_{i=1}^{n}\alpha'_i v_i$ where $\alpha'_i = \sum_{j=1}^{n} P_{ij}\alpha_j$ for each $i = 1, 2, \cdots, n$.

$$\begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1n} \\ P_{21} & P_{22} & \cdots & P_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n1} & P_{n2} & \cdots & P_{nn} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = \begin{bmatrix} \alpha'_1 \\ \alpha'_2 \\ \vdots \\ \alpha'_n \end{bmatrix} \tag{75}$$

$$\left[[w_1]_{\mathcal{B}_1} \mid [w_2]_{\mathcal{B}_1} \mid \cdots \mid [w_n]_{\mathcal{B}_1}\right][v]_{\mathcal{B}_2} = [v]_{\mathcal{B}_1}[v]_{\mathcal{B}_2} = [v]_{\mathcal{B}_1} \tag{76}$$

$$[id]_{\mathcal{B}_1}^{\mathcal{B}_2}[v]_{\mathcal{B}_2} = [v]_{\mathcal{B}_1} \tag{77}$$

$[id]_{\mathcal{B}_1}^{\mathcal{B}_2}$ can also weritten as $C(\mathcal{B}_2, \mathcal{B}_1)$.

15.4.2 Definition of change of basis matrix

The matrix $C(\mathcal{B}_2, \mathcal{B}_1) = [id]_{\mathcal{B}_1}^{\mathcal{B}_2}$ is called the change of basis matrix from $\mathcal{B}_2$ to $\mathcal{B}_1$.
Notice that $C(\mathcal{B}_2, \mathcal{B}_1)$ is invertible: $(C(\mathcal{B}_2, \mathcal{B}_1))^{-1} = C(\mathcal{B}_1, \mathcal{B}_2)$. That is $([id]_{\mathcal{B}_1}^{\mathcal{B}_2})^{-1} = [id]_{\mathcal{B}_2}^{\mathcal{B}_1}$.

15.4.3 Theorem of change of basis (unique invertible matrix)

Let V be an n-d.v.s. over $\mathbb{F}$ and let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two bases for V. There exists a unique necessary invertible $n \times n$ matrix A with entries in $\mathbb{F}$ s.t.

- $[v]_{\mathcal{B}_1} = A[v]_{\mathcal{B}_2}$
- $[v]_{\mathcal{B}_2} = A^{-1}[v]_{\mathcal{B}_1}$

  $\forall v \in V \ (A = C(\mathcal{B}_2, \mathcal{B}_1) = [id]_{\mathcal{B}_1}^{\mathcal{B}_2})$

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15.4.4 Theorem of change of basis (unique basis)

Suppose A is an invertible matrix in $\mathbb{F}^{n\times n}$. Let V be an n-d.v.s. over $\mathbb{F}$, and let $\mathcal{B}_1$ be an ordered basis for V. Then there is a unique basis $\mathcal{B}_2$ of V s.t.

- $[v]_{\mathcal{B}_1} = A[v]_{\mathcal{B}_2}$
- $[v]_{\mathcal{B}_2} = A^{-1}[v]_{\mathcal{B}_1}$

15.4.5 Proposition: composition of change of basis matrix

Let V be an n-d.v.s. over $\mathbb{F}$. Let $\mathcal{B}_1$, $\mathcal{B}_2$, $\mathcal{B}_3$ be three ordered bases for V. Then

$$[id]_{\mathcal{B}_3}^{\mathcal{B}_1} = [id]_{\mathcal{B}_3}^{\mathcal{B}_2}[id]_{\mathcal{B}_2}^{\mathcal{B}_1} \equiv C(\mathcal{B}_1, \mathcal{B}_3) = C(\mathcal{B}_2, \mathcal{B}_3)C(\mathcal{B}_1, \mathcal{B}_2) \tag{78}$$

15.4.6 Theorem: change of basis for matrix representation of l.t.

Let V and W be two f.d.v.s. over $\mathbb{F}$. Let B_1 and B_2 be two ordered bases for V and, let B'_1 and B'_2 be two ordered bases for W. If $T : V \to W$ is a l.t., then

$$[T]_{B'_2}^{B_2} = [id]_{B'_2}^{B'_1}[T]_{B'_1}^{B_1}[id]_{B_1}^{B_2} \equiv [T]_{B'_2}^{B_2} = C(B'_1, B'_2)[T]_{B'_1}^{B_1}C(B_2, B_1) \tag{79}$$

Proof see [104016 L-251208]-P17.

15.4.7 Corollary: change of basis for matrix representation of endomorphism

Let V be a f.d.v.s. over $\mathbb{F}$, and let B_1 and B_2 be two ordered bases for V. If T is an endomorphism (or linear operator $\equiv T : V \to V$ l.t.)
Then $[T]_{B_2} = [id]_{B_2}^{B_1}[T]_{B_1}[id]_{B_1}^{B_2} \equiv [T]_{B_2} = C^{-1}(B_2, B_1)[T]_{B_1}C(B_2, B_1)$.

15.4.8 Explaination of change of basis for matrix representation

Two matrices are similar if they represent the same linear transformation but with respect to different coordinate systems (bases). Geometrically, the transformation itself does not change—only the way we describe it in coordinates does.

15.5 Similar Matrices

15.5.1 Definition of similar matrices

Let A and B be $n \times n$ matrices with entries in $\mathbb{F}$. We say that B is similar to A if there exists an invertible matrix C s.t. $B = C^{-1}AC$.

15.5.2 Extension: similarity is a equivalence relation

Similarity is an equivalence relation.

i Reflexive: $A \sim_{sim} A$.

ii Symmetric: If $A \sim_{sim} B$, then $B \sim_{sim} A$.

iii Transitive: If $A \sim_{sim} B$ and $B \sim_{sim} C$, then $A \sim_{sim} C$.

15.5.3 Example

Example see [104016 L-251208]-P21.

16 Polynomial Vector Space and Lagrange Interpolation

16.1 Polynomial vector space

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16.1.1 Definition of polynomial vector space

Let $\mathbb{F}$ be a field and let $n \in \mathbb{N}_0$. The set of all polynomials of degree $\leq n$ with coefficients in $\mathbb{F}$ is denoted by $\mathbb{F}_n[x]$:

$$\mathbb{F}_n[x] = \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \mid a_0, a_1, \ldots, a_n \in \mathbb{F}\} \tag{80}$$

Together with polynomial addition and scalar multiplication, $\mathbb{F}_n[x]$ forms a vector space over $\mathbb{F}$.

16.1.2 Theorem: dimension of polynomial vector space

Let $\mathbb{F}_n[x]$ be the polynomial vector space over $\mathbb{F}$. Then

$$\dim_{\mathbb{F}} \mathbb{F}_n[x] = n + 1 \tag{81}$$

16.1.3 Standard basis of polynomial vector space

The set $\varepsilon = \{1, x, x^2, \ldots, x^n\}$ is a basis for $\mathbb{F}_n[x]$, called the **standard basis**. In particular, every polynomial $p(x) = a_0 + a_1x + \cdots + a_nx^n$ has the coordinate vector

$$[p(x)]_\varepsilon = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{F}^{n+1} \tag{82}$$

16.1.4 Example: isomorphism between polynomial space and $\mathbb{F}^{n+1}$

The map $T : \mathbb{F}_n[x] \to \mathbb{F}^{n+1}$ defined by

$$T(a_0 + a_1x + \cdots + a_nx^n) = (a_0, a_1, \ldots, a_n) \tag{83}$$

is an isomorphism. Therefore $\mathbb{F}_n[x] \cong \mathbb{F}^{n+1}$.

16.2 Evaluation functionals

16.2.1 Definition of evaluation functional

Let $c \in \mathbb{F}$. The **evaluation functional** at c is the map $L_c : \mathbb{F}_n[x] \to \mathbb{F}$ defined by

$$L_c(p) = p(c) \tag{84}$$

16.2.2 Proposition: evaluation functional is a linear transformation

For any $c \in \mathbb{F}$, $L_c : \mathbb{F}_n[x] \to \mathbb{F}$ is a l.t. (i.e., a linear functional). For all $p, q \in \mathbb{F}_n[x]$ and $\alpha \in \mathbb{F}$:

i $L_c(p+q) = (p+q)(c) = p(c) + q(c) = L_c(p) + L_c(q)$

ii $L_c(\alpha p) = (\alpha p)(c) = \alpha p(c) = \alpha L_c(p)$

16.2.3 Remark: family of evaluation functionals

Given distinct points $c_0, c_1, \ldots, c_n \in \mathbb{F}$, the map

$$T : \mathbb{F}_n[x] \to \mathbb{F}^{n+1}, \quad p \mapsto (p(c_0), p(c_1), \ldots, p(c_n)) \tag{85}$$

is a l.t.. As shown by the interpolation theorem below, this map is an isomorphism when the c_i are pairwise distinct.

16.3 The interpolation problem

16.3.1 Statement of the interpolation problem

Given $n+1$ distinct points $c_0, c_1, \ldots, c_n \in \mathbb{F}$ and prescribed values $y_0, y_1, \ldots, y_n \in \mathbb{F}$, the **interpolation problem** asks: does there exist a polynomial $p(x) \in \mathbb{F}_n[x]$ s.t.

$$p(c_i) = y_i \quad \forall i = 0, 1, \ldots, n \tag{86}$$

In other words, we seek $p(x)$ of degree $\leq n$ passing through all given data points $(c_0, y_0), (c_1, y_1), \ldots, (c_n, y_n)$.

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16.3.2 Reformulation as a linear map

Define the evaluation map

$$T : \mathbb{F}_n[x] \to \mathbb{F}^{n+1}, \quad p(x) \mapsto (p(c_0), p(c_1), \ldots, p(c_n)) \tag{87}$$

Then the interpolation problem is equivalent to: given $y = (y_0, y_1, \ldots, y_n) \in \mathbb{F}^{n+1}$, find $p(x) \in \mathbb{F}_n[x]$ s.t. $T(p) = y$.

16.3.3 Theorem: unique existence of interpolating polynomial

If $c_0, c_1, \ldots, c_n \in \mathbb{F}$ are pairwise distinct, then for any $y_0, y_1, \ldots, y_n \in \mathbb{F}$, there exists a **unique** polynomial $p(x) \in \mathbb{F}_n[x]$ s.t. $p(c_i) = y_i$ for all $i = 0, 1, \ldots, n$.

Equivalently, the evaluation map $T : \mathbb{F}_n[x] \to \mathbb{F}^{n+1}$ is an isomorphism.

16.4 Lagrange interpolation

16.4.1 Lagrange basis polynomials

Given pairwise distinct points $c_0, c_1, \ldots, c_n \in \mathbb{F}$, the **Lagrange basis polynomials** are defined by

$$\ell_i(x) = \prod_{j \neq i} \frac{x - c_j}{c_i - c_j} \tag{88}$$

for $i = 0, 1, \ldots, n$. Each $\ell_i(x) \in \mathbb{F}_n[x]$ and satisfies the **Kronecker delta property**:

$$\ell_i(c_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \tag{89}$$

16.4.2 Remark: Lagrange basis is a basis for $\mathbb{F}_n[x]$

The set $\{\ell_0(x), \ell_1(x), \ldots, \ell_n(x)\}$ is a basis for $\mathbb{F}_n[x]$, called the **Lagrange basis** associated to the points $c_0, c_1, \ldots, c_n$. Each $\ell_i(x)$ has degree at most n, so the set lies in $\mathbb{F}_n[x]$. Since there are $n+1$ such polynomials and $\dim \mathbb{F}_n[x] = n+1$, it suffices to show they are l.i.. Suppose $\sum_{i=0}^n \alpha_i \ell_i(x) = 0$. Evaluating at c_j and using the Kronecker delta property $\ell_i(c_j) = \delta_{ij}$ gives $\alpha_j = 0$ for each $j = 0, 1, \ldots, n$. Hence they are l.i. and form a basis.

16.4.3 Theorem: Lagrange interpolation formula

Let $c_0, c_1, \ldots, c_n \in \mathbb{F}$ be pairwise distinct and let $y_0, y_1, \ldots, y_n \in \mathbb{F}$. The unique interpolating polynomial $p(x) \in \mathbb{F}_n[x]$ s.t. $p(c_i) = y_i$ for all i is given by the **Lagrange interpolation formula**:

$$p(x) = \sum_{i=0}^n y_i \, \ell_i(x) = \sum_{i=0}^n y_i \prod_{j \neq i} \frac{x - c_j}{c_i - c_j} \tag{90}$$

This is simply the coordinate representation of $p(x)$ in the Lagrange basis: $p(x) = \sum_{i=0}^n y_i \ell_i(x)$ with coordinates $(y_0, y_1, \ldots, y_n)$.

16.4.4 Example: Lagrange interpolation with three data points

Find $p(x) \in \mathbb{R}_2[x]$ s.t. $p(1) = 2$, $p(2) = 3$, $p(3) = 5$.
Set $c_0 = 1$, $c_1 = 2$, $c_2 = 3$ and $y_0 = 2$, $y_1 = 3$, $y_2 = 5$. The Lagrange basis polynomials are:

$$\begin{aligned}
\ell_0(x) &= \frac{(x-2)(x-3)}{(1-2)(1-3)} = \frac{(x-2)(x-3)}{2} \\
\ell_1(x) &= \frac{(x-1)(x-3)}{(2-1)(2-3)} = \frac{(x-1)(x-3)}{-1} = -(x-1)(x-3) \\
\ell_2(x) &= \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{(x-1)(x-2)}{2}
\end{aligned}$$

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Applying the Lagrange formula and expanding each term:

$$\begin{aligned}
2\,\ell_0(x) &= (x-2)(x-3) = x^2 - 5x + 6 \\
3\,\ell_1(x) &= -3(x-1)(x-3) = -3x^2 + 12x - 9 \\
5\,\ell_2(x) &= \tfrac{5}{2}(x-1)(x-2) = \tfrac{5}{2}x^2 - \tfrac{15}{2}x + 5
\end{aligned}$$

Summing the coefficients:

$$\begin{aligned}
[x^2]: &\quad 1 - 3 + \tfrac{5}{2} = \tfrac{1}{2} \\
[x^1]: &\quad -5 + 12 - \tfrac{15}{2} = 7 - \tfrac{15}{2} = -\tfrac{1}{2} \\
[x^0]: &\quad 6 - 9 + 5 = 2
\end{aligned}$$

Therefore:

$$p(x) = \frac{1}{2}x^2 - \frac{1}{2}x + 2 \tag{91}$$

Verification: $p(1) = \frac{1}{2} - \frac{1}{2} + 2 = 2$ ✓, $p(2) = 2 - 1 + 2 = 3$ ✓, $p(3) = \frac{9}{2} - \frac{3}{2} + 2 = 3 + 2 = 5$ ✓.

17 Dual Space and Double Dual

17.1 Linear functionals

17.1.1 Definition of linear functional

Let V be a v.s. over a field $\mathbb{F}$. A **linear functional** on V is a l.t.

$$f : V \to \mathbb{F} \tag{92}$$

i.e. a map s.t. $f(u+v) = f(u) + f(v)$ and $f(\alpha v) = \alpha f(v)$ for all $u, v \in V$, $\alpha \in \mathbb{F}$.

17.1.2 Examples of linear functionals

i $f : \mathbb{R}^n \to \mathbb{R}$ defined by $f(x_1, x_2, \ldots, x_n) = a_1x_1 + a_2x_2 + \cdots + a_nx_n$ for fixed $a_i \in \mathbb{R}$.

ii $f : \mathbb{F}_n[x] \to \mathbb{F}$ defined by $f(p) = p(c)$ for a fixed $c \in \mathbb{F}$ (evaluation functional).

iii $f : \mathbb{F}[x] \to \mathbb{R}$ defined by $f(p) = \int_a^b p(t)\,dt$ for fixed $a, b \in \mathbb{R}$ (when $\mathbb{F} = \mathbb{R}$).

iv $f : M_{n\times n}(\mathbb{F}) \to \mathbb{F}$ defined by $f(A) = \operatorname{tr}(A)$ (trace functional).

17.2 Dual space

17.2.1 Definition of dual space

Let V be a v.s. over $\mathbb{F}$. The **dual space** of V is the set of all linear functionals on V:

$$V^* = \operatorname{Hom}_{\mathbb{F}}(V, \mathbb{F}) = \{f : V \to \mathbb{F} \mid f \text{ is a l.t.}\} \tag{93}$$

V^* is itself a v.s. over $\mathbb{F}$ under pointwise addition and scalar multiplication:

- $(f + g)(v) = f(v) + g(v)$
- $(\alpha f)(v) = \alpha f(v)$

17.2.2 Theorem: dimension of dual space

If V is a f.d.v.s. over $\mathbb{F}$, then

$$\dim_{\mathbb{F}} V^* = \dim_{\mathbb{F}} V \tag{94}$$

In particular, $V^* \cong V$ (though not canonically).

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17.2.3 Definition of dual basis

Let V be a f.d.v.s. over $\mathbb{F}$ with ordered basis $B = \{v_1, v_2, \ldots, v_n\}$. The **dual basis** of B is the set $B^* = \{f_1, f_2, \ldots, f_n\} \subseteq V^*$ defined by

$$f_i(v_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \tag{95}$$

for $i, j = 1, 2, \ldots, n$.

17.2.4 Theorem: dual basis is a basis for V^*

The dual basis $B^* = \{f_1, f_2, \ldots, f_n\}$ is a basis for V^*.

Proof (well-definedness). Each f_i is well-defined as a l.t. by the theorem of unique transformation (see Section 14.6), since we prescribe its values on the basis B.

Proof (linear independence). Suppose $\sum_{i=1}^n \alpha_i f_i = 0$. For each j, evaluating at v_j gives $\alpha_j = \left(\sum_{i=1}^n \alpha_i f_i\right)(v_j) = \sum_{i=1}^n \alpha_i \delta_{ij} = \alpha_j = 0$. Hence all $\alpha_j = 0$.

Proof (spanning). Let $f \in V^*$ and set $\alpha_i = f(v_i)$ for each i. Then for each j: $\left(\sum_{i=1}^n \alpha_i f_i\right)(v_j) = \alpha_j = f(v_j)$. Since both f and $\sum_{i=1}^n \alpha_i f_i$ agree on every basis vector v_j, by the theorem of unique transformation $f = \sum_{i=1}^n f(v_i) f_i$. $\square$

17.2.5 Remark: coordinate formula via dual basis

If $v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n \in V$, then for each i:

$$f_i(v) = \alpha_i \tag{96}$$

So the dual basis functionals **read off the coordinates** of a vector relative to B.

17.2.6 Example: dual basis of $\mathbb{R}^2$

Let $V = \mathbb{R}^2$ with standard basis $B = \{e_1, e_2\} = \{(1,0),(0,1)\}$. The dual basis is $B^* = \{f_1, f_2\}$ where

$$\begin{aligned} f_1(x_1,x_2) &= x_1 \quad \text{(projection onto the first coordinate)} \\ f_2(x_1,x_2) &= x_2 \quad \text{(projection onto the second coordinate)} \end{aligned}$$

More generally, with basis $B = \{(1,1),(1,-1)\}$, one checks that

$$\begin{aligned} f_1(x_1,x_2) &= \tfrac{x_1+x_2}{2} \\ f_2(x_1,x_2) &= \tfrac{x_1-x_2}{2} \end{aligned}$$

satisfies $f_i(v_j) = \delta_{ij}$.

17.3 Double dual

17.3.1 Definition of double dual

The **double dual** (or second dual) of V is the dual space of V^*:

$$V^{**} = (V^*)^* = \mathrm{Hom}_{\mathbb{F}}(V^*, \mathbb{F}) \tag{97}$$

17.3.2 Definition of canonical embedding

There is a natural l.t. $\varphi : V \to V^{**}$ called the **canonical embedding**, defined by

$$\varphi(v)(f) = f(v) \quad \forall v \in V,\ f \in V^* \tag{98}$$

That is, each vector $v \in V$ is sent to the "evaluation at v" functional $\hat{v} = \varphi(v) : V^* \to \mathbb{F}$, $f \mapsto f(v)$.

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17.3.3 Proposition: canonical embedding is a monomorphism

The canonical embedding $\varphi : V \to V^{**}$ is always injective (monomorphic), i.e. $\mathrm{Ker}\,\varphi = \{0\}$.

Proof. Suppose $\varphi(v) = 0$, i.e. $f(v) = 0$ for all $f \in V^*$. If $v \neq 0$, one can extend $\{v\}$ to a basis $\{v, v_2, \ldots, v_n\}$ of V (by the basis extension theorem, see Section 10.11) and define $f \in V^*$ by $f(v) = 1$ and $f(v_i) = 0$ for $i \geq 2$ (by the theorem of unique transformation, see Section 14.6). This gives $f(v) = 1 \neq 0$, a contradiction. Hence $v = 0$. □

17.3.4 Theorem: finite-dimensional spaces are canonically isomorphic to their double dual

If V is a f.d.v.s. over $\mathbb{F}$, then the canonical embedding $\varphi : V \to V^{**}$ is an isomorphism:

$$V \cong V^{**} \tag{99}$$

Proof. We have $\dim V^{**} = \dim V^* = \dim V$. Since φ is injective (monomorphism) and $\dim V = \dim V^{**}$, by the rank-nullity theorem (see Section 13.3) φ is also surjective (epimorphism). Hence φ is an isomorphism. □

17.3.5 Remark: canonical vs. non-canonical isomorphism

Although $V \cong V^*$ and $V \cong V^{**}$ both hold when $\dim V < \infty$, the isomorphisms are fundamentally different:

i $V \cong V^*$: this isomorphism depends on the choice of basis (not canonical).

ii $V \cong V^{**}$: the canonical embedding φ is basis-independent (canonical). This means V and V^{**} can be identified naturally without making any choices.

17.4 Transpose of a linear transformation

17.4.1 Definition of transpose (dual map)

Let $T : V \to W$ be a l.t.. The **transpose** (or **dual map**) of T is the l.t.

$$T^t : W^* \to V^* \tag{100}$$

defined by

$$T^t(g) = g \circ T \quad \forall g \in W^* \tag{101}$$

That is, $(T^t(g))(v) = g(T(v))$ for all $v \in V$.

17.4.2 Proposition: transpose is a linear transformation

$T^t : W^* \to V^*$ is a l.t.. For all $g, h \in W^*$ and $\alpha \in \mathbb{F}$:

i $T^t(g + h) = (g + h) \circ T = g \circ T + h \circ T = T^t(g) + T^t(h)$

ii $T^t(\alpha g) = (\alpha g) \circ T = \alpha(g \circ T) = \alpha T^t(g)$

17.4.3 Theorem: matrix of the transpose

Let V and W be f.d.v.s. over $\mathbb{F}$ with ordered bases $B = \{v_1, \ldots, v_n\}$ and $B' = \{w_1, \ldots, w_m\}$, and let $B^* = \{f_1, \ldots, f_n\}$, $B'^* = \{g_1, \ldots, g_m\}$ be the respective dual bases. If $A = [T]_{B'}^{B}$ is the $m \times n$ matrix of T, then the matrix of T^t with respect to B'^* and B^* is the **transpose matrix** A^T:

$$[T^t]_{B^*}^{B'^*} = A^T = ([T]_{B'}^{B})^T \tag{102}$$

17.4.4 Proposition: rank of transpose equals rank of T

For a l.t. $T : V \to W$ between f.d.v.s.,

$$\operatorname{rank}(T^t) = \operatorname{rank}(T) \tag{103}$$

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17.4.5 Example: transpose of a matrix transformation

Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be defined by the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$, i.e. $T(x) = Ax$.

The transpose map $T^t : (\mathbb{R}^3)^* \to (\mathbb{R}^2)^*$ is given, under identification of $(\mathbb{R}^n)^*$ with $\mathbb{R}^n$ via the standard dual basis, by the matrix

$$A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \tag{104}$$

So $(T^t(g))(x) = g(Ax)$, which corresponds to applying the row vector $g^T A$ to x.

18 Determinant

18.1 Introduction

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible iff $ad - cb \neq 0$. We can prove that by write this matrix into row reduced echelon form.

18.2 Definition of multilinear and alternating function

Let $A = (A_1 \mid A_2 \mid \cdots \mid A_n)$ be in $\mathbb{F}^{n\times n}$. A function $f : \mathbb{F}^{n\times n} \to \mathbb{F}$ is:
[i] Multilinear (in columns) if
$f(A_1 \mid \cdots \mid \alpha A_i + A_i' \mid \cdots \mid A_n) = \alpha f(A_1 \mid \cdots \mid A_i \mid \cdots \mid A_n) + f(A_1 \mid \cdots \mid A_i' \mid \cdots \mid A_n)$ for all $\alpha \in \mathbb{F}$, $A_i, A_i' \in \mathbb{F}^n$ and $1 \leq i \leq n$.
[ii] Alternating (in columns) if
$f(A_1 \mid \cdots \mid A_i \mid \cdots \mid A_j \mid \cdots \mid A_n) = 0$ if $A_i = A_j$ for some $i \neq j$.
If f is alternating multilinear (or form) and $f(I_n) \neq 0$, we say that f is a volume form.

18.3 Definition of determinant

The unique alternating multilinear function $f : \mathbb{F}^{n\times n} \to \mathbb{F}$ such that $f(I_n) = 1$ is denoted by det, so $\det(A)$ denotes the determinant of A.
Notation: $\det(A) = |A|$

18.4 Definition: determinant of linear transformation under different basis

$\det(f) = \det([f]_B)$ for some B basis for V.

18.5 Remark: one linear transformation has same determinant under different basis

We can prove that a l.t. has same determinant regardless of the choice of basis.
Proof see [104016 L-251217]-P15.

18.6 Proposition: relation between alt. mult. function and determinant

Let $f : \mathbb{F}^{n\times n} \to \mathbb{F}$ be an alternating multilinear function s.t. $f(Id) = \alpha$. Then $f = \alpha \cdot \det$.
Proof see [104016 L-251215]-P15.

18.7 Corollary: the existence of determinant

There exists a determinant from $\mathbb{F}^{n\times n} \to \mathbb{F}$ for each n in $\mathbb{N}$.

18.8 Theorem: uniqueness of alt. mult. function

Let $\alpha \in \mathbb{F}$. For each $n \in \mathbb{N}$, there exists a unique alternating multilinear function such that $f(I_n) = \alpha$.

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18.9 Corollary: uniqueness of determinant

For each $n \in \mathbb{N}$, the determinant is unique ($\alpha = 1$).

18.10 Example of determinant function for 2×2 matrix

$f : \mathbb{F}^{2\times 2} \to \mathbb{F}$ given by $f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$ is a determinant.
Proof see [104016 L-251210]-P5.

18.11 Proposition for alt. mult. function

If f is an alternating multilinear function ($f : \mathbb{F}^{n\times n} \to \mathbb{F}$, f is a form). Then,
(a) $f(A_1 \mid \cdots \mid A_i \mid \cdots \mid A_j \mid \cdots \mid A_n) = -f(A_1 \mid \cdots \mid A_j \mid \cdots \mid A_i \mid \cdots \mid A_n)$.
(b) $f(A_1 \mid \cdots \mid A_{i-1} \mid 0 \mid A_{i+1} \mid \cdots \mid A_n) = 0$
(c) $f(A_1 \mid \cdots \mid A_i + \alpha A_j \mid \cdots \mid A_j \mid \cdots \mid A_n) = f(A_1 \mid \cdots \mid A_i \mid \cdots \mid A_j \mid \cdots \mid A_n)$ for all $\alpha \in \mathbb{F}$.
(d) If $A_i = \sum\limits_{j=1\&j\neq i}^{n} \alpha_i A_j$, then $f(A_1 \mid \cdots \mid A_i \mid \cdots \mid A_n) = 0$.
Proof see [104016 L-251210]-P7.

18.12 Corollary: rank of a matrix and alt. mult. function

If f is an alternating multilinear function and $A \in \mathbb{F}^{n\times n}$. With rank $A < n$, then $f(A) = 0$.

18.13 Examples of alt. mult. function

1 $f : \mathbb{F}^{1\times 1} \to \mathbb{F}$ is alternating multilinear iff f is a l.t.

2 $f : \mathbb{F}^{2\times 2} \to \mathbb{F}$, $f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$, is an alternating multilinear function.

3 f is an alternating multilinear function if $f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \alpha(ad - cb)$, with $f(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}) = \alpha$

18.14 Proposition: uniqueness of determinant for 2×2 matrix and it's formula

$f : \mathbb{F}^{2\times 2} \to \mathbb{F}$ given by $f(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) = ad - cb$ is the unique determinant from $\mathbb{F}^{2\times 2}$ to $\mathbb{F}$.
Proof see [104016 L-251210]-P15.

18.15 Lemma: alt. mult. function from $(n+1)\times(n+1)$ matrix to $n\times n$

Let $f : \mathbb{F}^{(n+1)\times(n+1)} \to \mathbb{F}$ be an alternating multilinear function with $f(I_{n+1}) = \alpha$. Let $f_i : \mathbb{F}^{n\times n} \to \mathbb{F}$ be the function defined by $f_i(A) = f(A_{(i)})$ where:

$$A_{(i)} = \begin{pmatrix} a_{1,1} & \cdots & a_{1,i-1} & 0 & a_{1,i} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i-1,1} & \cdots & a_{i-1,i-1} & 0 & a_{i-1,i} & \cdots & a_{i-1,n} \\ 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ a_{i,1} & \cdots & a_{i,i-1} & 0 & a_{i,i} & \cdots & a_{i,n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,i-1} & 0 & a_{n,i} & \cdots & a_{n,n} \end{pmatrix}, \text{with } A = (a_{ij}) \in \mathbb{F}^{n\times n} \text{ for } i = 1, 2, \cdots, n \tag{105}$$

Then f_i is an alternating multilinear function.

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18.16 Formular for 2×2 and 3×3 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21} \tag{106}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \text{ (by rows)} \tag{107}$$

$$= a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{21}\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{31}\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \text{ (by cols)} \tag{108}$$

18.17 Notation of matrix without certain row and col

Given $A \in \mathbb{F}^{n\times n}$, $1 \le i, j \le n$.
$A(i \mid j) \in \mathbb{F}^{(n-1)\times(n-1)}$ is the matrix A without the i-th row and j-th column.

18.18 Proposition: alt. mult. function from $n\times n$ matrix to $(n-1)\times(n-1)$

Let $g : \mathbb{F}^{(n-1)\times(n-1)} \to \mathbb{F}$ be an alternating multilinear function with $g(I) = \alpha$ and $n \ge 2$. Then the function $f : \mathbb{F}^{n\times n} \to \mathbb{F}$ defined by

$$f(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} g(A(i \mid j)) \text{ is an alt. mult. funct. with } f(I_n) = \alpha \tag{109}$$

18.19 Proposition: determinant expansion by row or column

Given $A \in \mathbb{F}^{n\times n}$

$$\det(A) = \sum_{i=1}^{n}(-1)^{i+j}a_{ij}\det(A(i \mid j)) = \sum_{j=1}^{n}(-1)^{i+j}a_{ij}\det(A(i \mid j)) \tag{110}$$

The former one is using j-th column, the latter one is using i-th row.
Example see [104016 L-251215]-P5 and [104016 L-251215]-P8.
Remark: while calculating determinant, we can choose any row or column to expand, therefore, we need to choose the rows and cols with most 0 to simplify the calculation.

18.20 Proposition: determinant of transpose matrix

For all $A \in \mathbb{F}^{n\times n}$, $\det(A) = \det(A^t)$.
Proof by induction, see [104016 L-251215]-P9.

18.21 Proposition: elementary row operations and determinant

Let $A \in \mathbb{F}^{n\times n}$.
$\det(A') = \alpha \det A$ if $A' = M_i(\alpha)A$. $r_i \mapsto \alpha r_i$. First type of elementary row operation.
$\det(A') = \det A$ if $A' = L_i(j,\alpha)(A)$. $r_i \mapsto r_i + \alpha r_j$. Second type of elementary row operation.
$\det(A') = -\det A$ if $A' = T_{i,j}(A)$. $r_i \leftrightarrow r_j$. Third type of elementary row operation.

18.22 Proposition: determinant of upper triangular matrix

Let $A = (a_{ij} \in \mathbb{F}^{n\times n})$ s.t. $a_{ij} = 0$ for all i, j s.t. $i > j$.

$$\det(A) = \prod_{i=1}^{n} a_{ii} \tag{111}$$

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Proof by induction, see [104016 L-251215]-P13.

18.23 Corollary: determinant of diagonal matrix

If $D = \operatorname{diag}(a_{11}, a_{22}, \cdots, a_{nn})$, then

$$\det(D) = \prod_{i=1}^{n} a_{ii} \tag{112}$$

18.24 Proposition: determinant of matrix product

Let $A, B \in \mathbb{F}^{n\times n}$. Then

$$\det(AB) = \det(A)\cdot\det(B) \tag{113}$$

Proof see [104016 L-251215]-P16.

18.25 Corollary: properties of determinant

- $\det(AB) = \det(BA)$.
- Let $A \in \mathbb{F}^{n\times n}$, B an invertible matrix in $\mathbb{F}^{n\times n}$. Then $\det(BAB^{-1}) = \det(A)$. (similar matrix have same determinant)
- Let $A \in \mathbb{F}^{n\times n}$, $\alpha \in \mathbb{F}$. Then $\det(\alpha A) = \alpha^n \det(A)$.

18.26 Proposition: invertibility and determinant

Let $A \in \mathbb{F}^{n\times n}$, A is invertible iff $\det A \neq 0$
Proof from two direction, see [104016 L-251217]-P1.

18.27 Corollary: determinant of inverse matrix

$$\det A^{-1} = \frac{1}{\det(A)} \tag{114}$$

Proof see [104016 L-251217]-P1.

18.28 Remark: determinant of sum of matrices

$$\det(A+B) \neq \det A + \det B \tag{115}$$

18.29 Example of determinant of sum of matrices

$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. $A + B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. $\det(A+B) = 0 \neq 2 = \det A + \det B$.

18.30 Adjoint

18.30.1 Definition of adjoint matrix

Given $A \in \mathbb{F}^{n\times n}$, the adjoint $\operatorname{adj} A$ of A is given by

$$(\operatorname{adj} A)_{ij} = (-1)^{i+j} \det(A(j \mid i)) \tag{116}$$

18.30.2 Proposition: relation between adjoint and inverse

Given $A \in \mathbb{F}^{n\times n}$, $A \operatorname{adj} A = \operatorname{adj} A \cdot A = \det A \cdot I_n$.
Thus, if A is invertible, then $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$.
Proof left as exercise, see Workshop-10. Why the product is commutative also need to be proved.

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18.30.3 Example of adjoint and inverse

Let $A = \begin{pmatrix} 1 & 5 & 2 \\ 2 & 1 & 1 \\ 1 & 2 & 0 \end{pmatrix}$
$(\operatorname{adj} A)_{13} = (-1)^{1+3} \det \begin{pmatrix} 5 & 2 \\ 1 & 1 \end{pmatrix} = 3$
The full calculation see [104016 L-251217]-P4.
And $\operatorname{adj} A = \begin{pmatrix} -2 & 4 & 3 \\ 1 & -2 & 3 \\ 3 & 3 & -9 \end{pmatrix}$
$A \cdot \operatorname{adj} A = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix}$, $\det A = 9$
We get $A \cdot \operatorname{adj} A = \det A \cdot I$
$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$

19 Diagonalization and eigenvalues

19.1 Aim of diagonalization

Aim: Given a n-dimensional v.s. V over $\mathbb{F}$ and a l.op. $f : V \to V$, find a ordered basis B for V s.t. $[f]_B = \operatorname{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n)$.

19.2 Definition of diagonalizable linear transformation

A linear transformation $f : V \to V$ is called diagonalizable if there exists an ordered basis B for V s.t. $[f]_B = \mathrm{diag}(\lambda_1, \lambda_2, \cdots, \lambda_n) = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \lambda_n \end{bmatrix}$.

19.3 Definition of eigenvector

We say that a nonzero vector v in V is an eigenvector if there exists a scalar $\lambda \in \mathbb{F}$ s.t.

$$f(v) = \lambda v \tag{117}$$

19.4 Definition of eigenvalue and eigenspace

A scalar $\lambda \in \mathbb{F}$ is called an eigenvalue of f if $\mathrm{Ker}(f - \lambda\,\mathrm{id}) \neq \{0\} \equiv \exists v \in V \backslash \{0\}$ s.t. $(f - \lambda\,\mathrm{id})(v) = 0$ $\equiv \exists v \in V \backslash \{0\} s.t. f(v) = \lambda v$
The set $V_\lambda = \{v \in V \mid f(v) = \lambda v\}$ is called the eigenspace of f associated with λ (or relative to λ).
The elements ($\neq 0$) are called eigenvectors of f corresponding to λ.

19.5 Remark: eigenspace is a subspace

Given a scalar λ in $\mathbb{F}$ and $f : V \to V$ a l.t. $V_\lambda = \{v \in V \mid f(v) = \lambda v\}$ is a subspace of V.

19.6 Remark: eigenspace as kernel

$f(v) = \lambda v \equiv f(v) - \lambda v = 0 \equiv f(v) - \lambda\,\mathrm{id}(v) = 0 \equiv (f - \lambda\,\mathrm{id})(v) = 0 \equiv v \in \mathrm{Ker}(f - \lambda\,\mathrm{id})$.

$$V_\lambda = \{v \in V \mid f(v) = \lambda v\} = \mathrm{Ker}(f - \lambda\,\mathrm{id}) \tag{118}$$

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19.7 Theorem: equivalent conditions for eigenvalue

The following are equivalent:

- λ is an eigenvalue
- $f - \lambda\,\mathrm{id}$ is not a monomorphism
- $\det(f - \lambda\,\mathrm{id}) = 0$

Proof see [104016 L-251217]-P13 and P15.

19.8 Characteristic polynomial

19.8.1 Definition of characteristic polynomial

Let $f : V \to V$ be a l.op. on a v.s. V over $\mathbb{F}$, and let B be an oredered basis for V. The monic polynomial $\det(x\,\mathrm{id} - [f]_B)$ is called the characteristic polynomial of f.

$$\chi_f(x) = \det(x\,\mathrm{id} - [f]_B) \tag{119}$$

19.8.2 Corollary: eigenvalues are the root of characteristic polynomial

$f : V \to V$ a l.op. on a f.d.v.s. V over $\mathbb{F}$. A scalar λ in $\mathbb{F}$ is an eigenvalue of f iff λ is a root of $\chi(x)$.
Proof see [104016 L-251217]-P16.

19.8.3 Remark: characteristic polynomial doesn't depend on basis

$\chi_f(x)$ doesn't depend on the choice of basis.
Proof see [104016 L-251217]-P17.

19.8.4 Example of characteristic polynomial and eigenvalues

Let $B = \{v_1, v_2\}$ be an oredered basis for $\mathbb{R}^2$ and let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be a l.op. given by $f(v_1) = 2v_1 + 2v_2$ and $f(v_2) = 3v_1 + v_2$.

$[f]_B = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$

Thus, $\chi_f(x) = \det \begin{pmatrix} x-2 & -3 \\ -2 & x-1 \end{pmatrix} = (x-2)(x-1) - 6 = x^2 - 3x - 4 = (x-4)(x+1)$

The characteristic polynomial of f is $\chi_f(x) = x^2 - 3x - 4 = (x-4)(x+1)$.

The eigenvalue are $\lambda_1 = 4$, $\lambda_2 = -1$.

$V_4 = \{v \in \mathbb{R}^2 \mid f(v) = 4v\} = \operatorname{Ker}(f - 4\,\mathrm{id})$

$\operatorname{Ker}(\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} - \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}) = \operatorname{Ker} \begin{pmatrix} -2 & 3 \\ 2 & -3 \end{pmatrix} = \operatorname{Ker} \begin{pmatrix} -2 & 3 \\ 0 & 0 \end{pmatrix} = \operatorname{span}\{(3, 2)\}$.

So, $V_4 = \operatorname{span}\{3v_1 + 2v_2\}$.

Similarly, $V_{-1} = \operatorname{span}\{v_1 - v_2\}$.

Conclusion: $\mathbb{R}^2 = V_4 \oplus V_{-1}$, and $B' = \{3v_1 + 2v_2, v_1 - v_2\}$ is an oredered basis for $\mathbb{R}^2$ s.t. $[f]_{B'} = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}$

$[f]_B = C(B', B)[f]_{B'}C(B, B')$

Where $C(B', B) = \begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix}$ and $C(B, B') = C(B', B)^{-1}$.

19.9 Proposition: intersection of eigenspaces

Let $f : V \to V$ be a l.op. on a f.d.v.s. V over $\mathbb{F}$. Let $\lambda_1 and \lambda_2$ be different eigenvalues of f. Then:

$$V_{\lambda_1} \cap V_{\lambda_2} = \{0\} \tag{120}$$

Proof see [104016 L-251222]-P15.

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19.10 Corollary: sum of two eigenspaces is direct

Let $f : V \to V$ be a l.op. on a f.d.v.s. V over $\mathbb{F}$. Let λ_1, λ_2 be different eigenvalues of f. Then: $V_{\lambda_1} + V_{\lambda_2} = V_{\lambda_1} \oplus V_{\lambda_2}$.

19.11 Theorem: sum of eigenspaces is direct

Let $f : V \to V$ be a l.op. on a f.d.v.s. V over $\mathbb{F}$.

Let $\lambda_1, \lambda_2, \cdots, \lambda_k$ be different eigenvalues for f. Then:

$V_{\lambda_1} + V_{\lambda_2} + \cdots + V_{\lambda_k} = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_k}$

19.12 Theorem: equivalent conditions for diagonalizability

Let $f : V \to V$ be a l.op. on a f.d.v.s. V over $\mathbb{F}$. Let $\lambda_1, \lambda_2, \cdots, \lambda_k$ be different eigenvalues for f and let V_{λ_i} be the eigenspace relative to the eigenvalue λ_i for $i = 1, 2, \cdots, k$. Then the following are equivalent

(i) f is diagonalizable.

(ii) The characteristic polynomial for f is

$$\chi_f(x) = \prod_{i=1}^{k} (x - \lambda_i)^{d_i} \tag{121}$$

and $\dim V_{\lambda_i} = d_i$ for $i = 1, 2, \cdots, k$.

(iii)

$$\dim V = \sum_{i=1}^{k} \dim V_{\lambda_i} \tag{122}$$

Note: d_i is the algebraic multiplicity of the eigenvalue λ_i. $d_i = \operatorname{mult}(\lambda_i, \chi_f(x))$. $\dim_{\mathbb{F}} V_{\lambda_i}$ is the geometric multiplicity of the eigenvalue λ_i.

Proof see [104016 L-251224]-P3.

19.13 Definition of geometric and algebraic multiplicity

(a) The number $k = \dim V_\lambda$ is called the geometric multiplicity.

(b) The multiplicity of λ in $\chi_f(x)$ is called the algebraic multiplicity.

19.14 Proposition: relation between geometric and algebraic multiplicity

Let f be a l.op. on a f.d.v.s. V over $\mathbb{F}$ and let λ be an eigenvalue of f. Then

$$1 \leq \dim V_\lambda \leq \operatorname{mult}(\lambda, \chi_f(x)) \tag{123}$$

Proof see [104016 L-251229]-P1.

19.15 Proposition: diagonalizability with n different eigenvalues

Let V be a n-d.v.s. over $\mathbb{F}$ and let f be a l.op. on V. If $\chi_f(x)$ has n different eigenvalues, then f is diagonalizable.

Proof see [104016 L-251229]-P4.

19.16 Examples of diagonalizable linear transformations and matrices

See [104016 L-251224]-P5. **Please review them.**

f is diagonalizable if $\exists$ a basis for V s.t. $[f]_{\mathcal{B}}$ is a diagonal matrix.

$A \in \mathbb{F}^{n\times n}$ is diagonalizable if $\exists$ an invertible matrix $P \in \mathbb{F}^{n\times n}$ s.t. $A = PDP^{-1}$ with D diagonal.

20 Annihilating Polynomials

20.1 Definition of polynomial of a linear operator

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Let V be a f.d.v.s. over $\mathbb{F}$ and f is a l.op. . Given $p(x) = \sum_{i=0}^{m} a_i x^i \in \mathbb{F}[x]$, then $P(f) = \sum_{i=0}^{m} a_i f^i$ where $f^0 = \mathrm{Id}_V$ and $f^i = f \circ f^{i-1}$ for $i \geq 1$ is a linear operator.

Suppose that $p(x)$, $q(x) \in \mathbb{F}[x]$ are s.t. $p(f) = \mathrm{O} = q(f)$. Then $(p+q)(f) = p(f) + q(f) = \mathrm{O}$.

Moreover, if $h(x) \in \mathbb{F}[x]$, then $(hp)(f) = h(f) \cdot p(f) = \mathrm{O}$.

Consider $L_f = \{p(x) \in \mathbb{F}[x] \mid p(f) = \mathrm{O}\}$

(i) $0 \in L_f$: $0(f)(v) = \mathrm{O}(v) = 0$

(ii) $p + q \in L_f$: $p, q \in L_f$, $(p+q)(f) = p(f) + q(f) = \mathrm{O}$

(iii) $hp \in L_f$: $p \in L_f$, $(h \cdot p)(f) = h(f)p(f) = \mathrm{O}$

(iv) $L_f \neq \mathbb{F}[x]$: Let $p(x) = 1$, then $p(f) = \mathrm{Id}_V$

Annihilating polynomial of f is a non-zero polynomial $p(x) \in \mathbb{F}[x]$ s.t. $p(f) = \mathrm{O}$.

20.2 Example of polynomial of l.op.

V a f.d.v.s. $p(x) = 2 + 6x + 7x^2 - 9x^3$, $q(x) = x - 1$

$p(\mathrm{id}) = 6\,\mathrm{id}$, $q(\mathrm{id}) = \mathrm{O}$

$q(x) = x - 1 \in L_{\mathrm{id}}$

$r(x) = x \notin L_{\mathrm{id}}$

20.3 Lemma: existence of annihilating polynomial

Let V ve a f.d.v.s. over $\mathbb{F}$ and let f be a l.op.. There exists a polynomial $p(x) \in \mathbb{F}[x]$, $p(x) \neq 0$ s.t. $p(f) = \mathrm{O}$.

Proof see [104016 L-251229]-P9.

20.4 Remark: minimal degree of annihilating polynomial

For $L_f \neq \{0\}$
Let $l = \min\{\deg(p(x)) \mid p(x) \in L_f \backslash \{0\}\}$. This minimum exists since $l \leq n^2$.

20.5 Proposition: existence and uniqueness of certain degree annihilating polynomial

Let V ve a f.d.v.s. over $\mathbb{F}$ and f be a l.op.
There exist a unique polynomial $m(x)$ in L of degree l. That is $m(f) = \mathrm{O}$ and $\deg m = l$.
Proof see [104016 L-251229]-P11

20.6 Definition of minimal polynomial

Let V be a f.d.v.s. over $\mathbb{F}$ and let f be a l.op. . The unique monic polynomial $m(x)$ s.t.

(i) $m(x)$ is a monic polynomial

(ii) $m(f) = \mathrm{O}$

(iii) $\deg m \leq \deg p(x)\ \forall p \in \mathbb{F}[x]$ s.t. $p(f) = \mathrm{O}$

is called the minimal polynomial of f.

20.7 Remark: properties of minimal polynomial

(1) In a similar way, the minimal polynomial of a matrix $A \in \mathbb{F}^{n\times n}$ is defined.

(2) Similar matrices have the same minimal polynomial. (Proof see [104016 L-251229]-P14)

(3) Ex: If $A = [f]_B$ where B is an ordered basis for V and f is a l.op. then $m_f(x) = m_A(x)$.

(4) If $p(x) \in \mathbb{F}[x]$, $p(f) = \mathrm{O}$ iff $m(x) \mid p(x)$. (Proof see [104016]-P16)

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20.8 Example of minimal polynomial

20.8.1 $f = \mathrm{id}$

$f = \mathrm{id} \Rightarrow m_f(x) = x + a_0$
$\mathrm{O} = m_f(f) = \mathrm{id} + a_0\,\mathrm{id} = (1 + a_0)\,\mathrm{id} \Rightarrow a_0 = -1$
Thus $m_f(x) = x - 1$.

20.8.2 $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, if $p(x) = x + a_0$, then $p(A) = A + a_0\,\mathrm{id} = \begin{bmatrix} a_0 & -1 \\ 1 & a_0 \end{bmatrix} \neq \mathrm{O}$
Thus, $\deg m_A \geq 2$.
$A^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -\,\mathrm{id}$
Thus, $A^2 + I = \mathrm{O}$ and $m_A(x) = x^2 + 1$.

20.9 Lemma: eigenvalue and minimal polynomial

Let V be a f.d.v.s. over $\mathbb{F}$ and let f be a l.op. on V. If $\lambda \in \mathbb{F}$, $v \in V$ s.t. $f(v) = \lambda v$, then for all $P \in \mathbb{F}[x]$, $P(f)(v) = P(\lambda)(v)$.
Proof see [104016 L-251231]-P3.

20.10 Theorem: minimal polynomial and characteristic polynomial share same roots

Let $f : V \to V$ be a l.op. on an n-dimensional v.s. V over $\mathbb{F}$. $\chi_f(x)$ and $m_f(x)$ have the same root.
Proof see [104016 L-251231]-P4.

20.11 Corollary: minimal polynomial of diagonalizable l.op.

If f is diagonalizable, with distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$, then (iff) $m_f(x) = (x-\lambda_1)(x-\lambda_2)\cdots(x-\lambda_k)$.

Proof see [104016 L-251231]-P7.

20.12 Corollary: inverse of invertible l.op. as polynomial of l.op.

Let V be a f.d.v.s. over $\mathbb{F}$ and $f : V \to V$ be a l.op.

If f is invertible, then there exists a polynomial $p(x) \in \mathbb{F}[x]$ s.t. $f^{-1} = p(f)$.

Proof see [104016 L-260105]-P13.

20.13 Cayley-Hamilton Theorem

20.13.1 Statement of Cayley-Hamilton Theorem

Let V be a f.d.v.s. over $\mathbb{F}$ and let f be a l.op. on V. Then

$$m_f(x) \mid \chi_f(x) \equiv \chi_f(f) = \mathrm{O} \tag{124}$$

20.13.2 Theorem: diagonalizability and minimal polynomial

Let V be a f.d.v.s. over $\mathbb{F}$ and let f be a l.op. on V. Then f is diagonalizable iff $m_f(x) = \prod_{i=1}^{k}(x-\lambda_i)$ where $\lambda_1, \lambda_2, \ldots, \lambda_k$ are distinct eigenvalues of f.

Proof see [104016 L-251231]-P10.

20.13.3 Example of non-diagonalizable matrix via minimal polynomial

$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$, $\chi_A(x) = (x-1)^2(x+1)^2(x-3)$, $m_A(x) = (x-1)^2(x+1)(x-3)$

$(A-I)(A+I)(A-3I) \neq \mathrm{O}$

$(A-I)^2(A+I)(A-3I) = \mathrm{O}$

A is not diagonalizable.

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20.14 Proof for Cayley-Hamilton Theorem

Let $f : V \to V$ be a l.op. on a f.d.v.s. V over $\mathbb{F}$. Then $m_f(x) \mid \chi_f(x) \equiv \chi_f(f) = \mathrm{O}$.

20.14.1 Lemma 1: cyclic vector and Cayley-Hamilton Theorem

Let V be a n-d.v.s. over $\mathbb{F}$ and let f be a l.op. on V. If there exist a vector $v \in V$ s.t. the set $\{v, f(v), f^2(v), \ldots, f^{n-1}(v)\}$ is a basis of V, then $\chi_f(f) = \mathrm{O}$.

Proof see [104016 L-260105]-P2.

20.14.2 Remark: polynomial of block matrix

If $A = \begin{bmatrix} A_1 & B \\ 0 & A_2 \end{bmatrix} \in \mathbb{F}^{n\times n}$ where $A_1 \in \mathbb{F}^{n_1\times n_1}$, $A_2 \in \mathbb{F}^{n_2\times n_2}$, and $n = n_1 + n_2$. Then for any polynomial $p(x) \in \mathbb{F}[x]$, we have $P(A) = \begin{bmatrix} P(A_1) & C \\ 0 & P(A_2) \end{bmatrix}$ for some matrix $C \in \mathbb{F}^{n_1\times n_2}$.

Let $p(x) = a_0 + a_1x + \cdots + a_mx^m \in \mathbb{F}[x]$. Then

$$\begin{aligned} P(A) &= a_0I_n + a_1A + a_2A^2 + \cdots + a_mA^m \\ &= \begin{bmatrix} a_0I_{n_1} & 0 \\ 0 & a_0I_{n_2} \end{bmatrix} + a_1\begin{bmatrix} A_1 & B \\ 0 & A_2 \end{bmatrix} + a_2\begin{bmatrix} A_1^2 & A_1B + BA_2 \\ 0 & A_2^2 \end{bmatrix} + \cdots + a_m\begin{bmatrix} A_1^m & * \\ 0 & A_2^m \end{bmatrix} \\ &= \begin{bmatrix} P(A_1) & C \\ 0 & P(A_2) \end{bmatrix} \end{aligned}$$

20.14.3 Lemma 2: block matrix and Cayley-Hamilton Theorem

Let $A = \begin{bmatrix} A_1 & B \\ 0 & A_2 \end{bmatrix}$ in $\mathbb{F}^{n\times n}$ where $A_1 \in \mathbb{F}^{n_1\times n_1}$, $A_2 \in \mathbb{F}^{n_2\times n_2}$, and $n = n_1 + n_2$. $\chi_{A_1}(A_1) = 0$ and $\chi_{A_2}(A_2) = 0$, then $\chi_A(A) = 0$.

Proof see [104016 L-260105]-P8.

20.14.4 Proof of Cayley-Hamilton Theorem

Let V be a n-d.v.s. over $\mathbb{F}$ and let f be a l.op. on V. Then $\chi_f(f) = \mathrm{O}$.

Proof see [104016 L-260105]-P9.

21 Proofs

21.1 Proof of the lemma in section 4.4.2

We know that there exist polynomials $r(x)$ and $t(x)$ in $\mathbb{F}[x]$ s.t.:

$$1 = r(x)f(x) + t(x)g(x)$$

Thus

$$h(x) = h(x)(r(x)f(x) + t(x)g(x)) = r(x)f(x)h(x) + t(x)g(x)h(x)$$

But $f(x)|r(x)f(x)h(x)$ and $f(x)|t(x)g(x)h(x)$, so

$$f(x)|r(x)f(x)h(x) + t(x)g(x)h(x) = h(x)$$

Q.E.D.

21.2 Proof of the proposition in section 8.4

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To prove that $S \cap T \leq V$, we need to prove the three conditions in the proposition of subspace:

i : $0 \in S \cap T$: $S \leq V$, $T \leq V$. $0 \in S$ and $0 \in T \Rightarrow 0 \in S \cap T$

ii : $u, v \in S \cap T$: $u, v \in S$ and $u, v \in T \Rightarrow u + v \in S$ and $u + v \in T \Rightarrow u + v \in S \cap T$

iii : $\alpha \in \mathbb{F}, v \in S \cap T \Rightarrow \alpha v \in S$ and $\alpha v \in T \Rightarrow \alpha v \in S \cap T$

Therefore $S \cap T \leq V$

21.3 Proof of the proposition in section 8.7

Let R be the row-echelon form that is row-equivalent to A by applying gauss elimination. Then $Ax = b$ has a solution for any $b \in \mathbb{F}^m$ iff $Rx = b'$ has solution for any $b' \in \mathbb{F}^m$.

Now, if R has a zero row, then we can always choose b' in $\mathbb{F}^m$, s.t. $Rx = b'$ is inconsistent, while if R has no zero rows then $Rx = b'$ can be solved no matter what b' is.

Then R has no zero rows is equivalent to say that each row has a leading entry, then $\mathrm{rank}(A) = m$.

In this case, $\mathrm{rank}(A) = m \leq n$, so $m \leq n$.

Q.E.D.

21.4 Proof of the proposition in section 9.8

Since $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = Ax$, the set $\{v_1, v_2, \cdots, v_n\}$ is l.i. iff $Ax = 0$ has only the trivial solution.

We already saw that this is the case if and only if $\mathrm{rank}(A) = n$,the number of columns in this case $n \leq m$.

Q.E.D.

21.5 Proof of the proposition in section 10.4

$v_1, v_2, \cdots, v_n$ in $\mathbb{F}^m$ are l.i. iff $\text{rank}(A) = n \leq m$.
And $\text{span}\{v_1, v_2, \cdots, v_n\} = \mathbb{F}^m$ iff $\text{rank}(A) = m \leq n$.
Therefore, $v_1, v_2, \cdots, v_n$ is a basis for $\mathbb{F}^m$ iff $n = m$ and $\text{rank}(A) = n$.
Q.E.D.

21.6 Proof of the theorem in section 10.5

21.6.1 (1)⇒(2)

(1)⇒(2): By definition of basis.

21.6.2 (2)⇒(3)

(2)⇒(3): If $v_1, v_2, \cdots, v_n$ are l.d., then there exist $j \in \{1, 2, \cdots, n\}$ s.t. $v_j = \sum\limits_{i \neq j, i=1}^{n} \alpha_i v_i$ and $v_j \in$ span$\{v_i \mid i \in \{1, 2, \cdots, n\} n \{j\}\}$.
But dim(span$\{v_i \mid i \in \{1, 2, \cdots, n\} n \{j\}\}$)≤n-1≠n=dim V, so V≠ span$\{v_1, v_2, \cdots,$

(3)⇒(1): If $v_1, v_2, \cdots, v_n$ are l.i., then the set $\{v_1, v_2, \cdots, v_n\}$ can be extended to a basis of V, but dim V=n=#$\{v_1, v_2, \cdots, v_n\}$.
So $\{v_1, v_2, \cdots, v_n\}$ is a basis for V.
Therefore (3)⇒(1).
Q.E.D.

21.7 Proof of the theorem in section 10.9

Choose a maximal subset of vectors $w_1, w_2, \cdots, w_r$ from $v_1, v_2, \cdots, v_n$ s.t.$w_i \notin \text{span} w_1, w_2, \cdots, w_{i-1}$
Then $w_1, w_2, \cdots, w_r$ is a basis of span$v_1, v_2, \cdots, v_n$:
if $\exists\ i \in \{1, 2, \cdots, n\}$ s.t. $v_i \notin \text{span}\{w_1, w_2, \cdots, w_r\}$
then $w_1, w_2, \cdots, w_r$ is not maximal.
Thus, span$v_1, v_2, \cdots, v_n \subseteq$ span$w_1, w_2, \cdots, w_r$. And $w_1, w_2, \cdots, w_r$ is a l.i. set.
Q.E.D.

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21.8 Proof of the proposition in section 10.10

Let $w_1, w_2, \cdots, w_n$ be the columns of R. Then $w_{i_1}, w_{i_2}, \cdots, w_{i_r}$ are the first canonical basis vectors $e_1, e_2, \cdots, e_r$ in $\mathbb{F}^m$, so they are l.i.
It's clear that $\text{span}\{w_{i_1}, w_{i_2}, \cdots, w_{i_r}\} = \text{span}\{w_1, w_2, \cdots, w_n\}$.
We prove that $\{v_{i_1}, v_{i_2}, \cdots, v_{i_r}\}$ is l.i. .:
Suppose that: $x_{i_1} v_1 + x_{i_2} v_2 + \cdots + x_{i_r} v_{i_r} = 0$
Define:

$$x_l = \begin{cases} 0 & \text{if } l \notin \{i_1, i_2, \cdots, i_r\} \\ x_{i_j} & \text{if } l = i_j \end{cases}$$

Then we get the vector x $\in \mathbb{F}^n$ s.t. $Ax = 0$.
Then Rx=0 and Rx=$x_1 w_1 + x_2 w_2 + \cdots + x_n w_n = x_{i_1} e_1 + x_{i_2} e_2 + \cdots + x_{i_r} e_r = 0$.
And then $x_{i_1} = x_{i_2} = \cdots = x_{i_r} = 0$. Thus $v_{i_1}, v_{i_2}, \cdots, v_{i_r}$ is l.i. set.
Now, $\{v_{i_1}, v_{i_2}, \cdots, v_{i_r}\}$ is a gererating set of span$\{v_1, v_2, \cdots, v_n\}$.
It's enough to show that for $l \notin \{i_1, i_2, \cdots, i_r\}$, $v_l \in \text{span}\{v_{i_1}, v_{i_2}, \cdots, v_{i_r}\}$.
Notice that $w_l = R_{1l} w_{i_1} + R_{2l} w_{i_2} + \cdots + R_{rl} w_{i_r}$. Where R_{ij} is the entry of R in the i-th row and j-th column.
Let $X \in \mathbb{F}^n$ be the vector whose l-th coordinate is 1, and whose i_k coordinate is $-R_{i_k l}$. Then:

$$Rx = w_l - R_{i_1 l} w_{i_1} - R_{i_2 l} w_{i_2} - \cdots - R_{i_r l} w_{i_r} = w_l - w_l = 0$$

Hence Ax=0. But Ax = $v_l - R_{i_1 l} v_{i_1} - R_{i_2 l} v_{i_2} - \cdots - R_{i_r l} v_{i_r}$=0.
So $v_{i_1}, v_{i_2}, \cdots, v_{i_r}$ is a basis for span$v_1, v_2, \cdots, v_n$.
Q.E.D.

21.9 Proof of the theorem in section 10.11

If $\{v_1, v_2, \cdots, v_k\}$ is a l.i. set if vectors with r¡dim$_{\mathbb{F}}$V, then there exists v_{r+1} in $V n$ span$\{v_1, v_2, \cdots, v_r\}$. Then $\{v_1, v_2, \cdots, v_r, v_{r+1}\}$ is a l.i. set.

Since dim $V = n$, the process must stop and there exist $n-r$ vectors $v_{r+1}, v_{r+2}, \cdots, v_n$ s.t. V span$\{v_1, v_2, \cdots, v_r, v_{r+1}, \cdots, v_n\}$.

Q.E.D.

21.10 Proof of the theorem in section 11.3

The vectors $w_1, w_2, \cdots, w_n$ are in span$v_1, v_2, \cdots, v_m$. If $n > m$ by the lemma $\{w_1, w_2, \cdots, w_n\}$ is l.d. , contradicting the fact that $\{w_1, w_2, \cdots, w_n\}$ is a basis.

Thus n≤m. By symmetry, m≤n. Hence $n = m$.

Q.E.D.

21.11 Proof of the theorem in section 11.9

Suppose the $W \neq \{0\}$ and let $n = \dim_{\mathbb{F}} V$.

By the fundamental lemma of linear independence, any l.i. set of vectors in W has at most n vectors.

Since $W \neq \{0\}$, there exists $w_i \in W n\{0\}$, then choose elements $w_1, w_2, \cdots, w_k$ in W s.t. $w_i \notin span\{w_1, w_2, \cdots, w_{i-1}\} \subseteq W$ and define $\beta_i = \{w_1, w_2, \cdots, w_i\}$.

Notice that β_i is a l.i. set of vectors in W and $1 \leq i \leq n$. Therefore, the process must stop after many steps, say after $w_1, w_2, \cdots, w_d$ with $d \leq n$ and $dim_{\mathbb{F}} W = d$. If there exists $w \in W n span\{w_1, w_2, \cdots, w_d\}$, then $\{w_1, w_2, \cdots, w_d, w\}$ is a l.i. set of vectors in W with d+1 vectors, contradicting the maximality of d.

Q.E.D.

21.12 Proof of proposition of the sum of subspaces is a subspace in section 12.3

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21.12.1 Property i

Since W_i is a subspace, $0 \in W_i \quad \forall i \in \{1, 2, \cdots, k\}$, then $0 + 0 + 0 + 0 + \cdots + 0 = 0 \in W$.

21.12.2 Property ii

If w and w' are in W, there exist $w_i, w_i' \in W_i$ s.t. $w = \sum w_i$ and $w' = \sum w_i'$.

Thus $w + w' = \sum\limits_{i=1}^{k} w_i + \sum\limits_{i=1}^{k} w_i' = \sum\limits_{i=1}^{k} (w_i + w_i')$. Since W_i is a subspace, $w_i + w_i' \in W_i$, so $w + w' \in W$.

21.12.3 Property iii

$\alpha \in \mathbb{F}$ and $w \in W$, $\alpha w = \alpha \sum\limits_{i=1}^{k} w_i = \sum\limits_{i=1}^{k} \alpha w_i$. Since W_i is a subspace, $\alpha w_i \in W_i$, so $\alpha w \in W$.

Therefore, W is a subspace of V. Q.E.D.

21.13 Proof of the Rank-Nullity Theorem in section 13.3

rank(A)=r=#non-zero rows of R=#pivot columns

dim Ker A=$n - r$=#free variables=#non-pivot columns

Therefore, rank(A)+dim Ker A=$r + (n - r) = n$.

Q.E.D.

21.14 Proof of theorem unique transformation in section 14.6

Define T(v_i)=w_i $\forall i = 1, 2, \cdots, n$.

Since $\{v_1, v_2, \cdots, v_n\}$ is a basis for V, for each v ∈ V, there exist unique scalars $\alpha_1, \alpha_2, \cdots, \alpha_n$ s.t.:

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$$

Thus, define:

$$T(v) = \alpha_1 w_1 + \alpha_2 w_2 + \cdots + \alpha_n w_n$$

T is a l.t.: if $v = \sum_{i=1}^{n} \alpha_i v_i$ and $u = \sum_{i=1}^{n} \beta_i v_i$,

T(u+v)=T($\sum_{i=1}^{n}(\alpha_i + \beta_i)v_i$)=$\sum_{i=1}^{n}(\alpha_i + \beta_i)w_i$=$\sum_{i=1}^{n} \alpha_i w_i + \sum_{i=1}^{n} \beta_i w_i$=T(u)+T(v).

T(αv)=T($\sum_{i=1}^{n} \alpha\alpha_i v_i$)=$\sum_{i=1}^{n} \alpha\alpha_i w_i$=$\alpha \sum_{i=1}^{n} \alpha_i w_i$=$\alpha$T(v).

Therefore, T is a l.t.

Now, if T' is a l.t. s.t. T'(v_i)=w_i $\forall i = 1, 2, \cdots, n$, then T'(v)=T'($\sum_{i=1}^{n} \alpha_i v_i$)=$\sum_{i=1}^{n} \alpha_i T'(v_i)$=$\sum_{i=1}^{n} \alpha_i w_i$=T(v).

Therefore T'=T.

Q.E.D.

21.15 Proof of theorem of the algebra of linear transformations in section 14.7

21.15.1 Addition

For $u, v \in V$,

$$\begin{aligned}(T_1 + T_2)(\alpha u + v) &= T_1(\alpha u + v) + T_2(\alpha u + v)\\ &= (T_1(\alpha u) + T_1(v)) + (T_2(\alpha u) + T_2(v))\\ &= (T_1(\alpha u) + T_2(\alpha u)) + (T_1(v) + T_2(v))\\ &= (T_1 + T_2)(\alpha u) + (T_1 + T_2)(v)\end{aligned}$$

21.15.2 Scalar multiplication

For $\alpha \in \mathbb{F}$ and $v \in V$,

$$\begin{aligned}(\alpha T_1)(\beta v + v') &= \alpha(T_1(\beta v + v'))\\ &= \alpha(T_1(\beta v) + T_1(v'))\\ &= \alpha T_1(\beta v) + \alpha T_1(v')\\ &= \beta(\alpha T_1)(v) + (\alpha T_1)(v')\end{aligned}$$

Q.E.D.

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21.16 Proof of proposition of composition of linear transformations in section 14.10

$$\begin{aligned}(T_2 \circ T_1)(\alpha u + v) &= T_2(T_1(\alpha u + v))\\ &= T_2(\alpha T_1(u) + T_1(v))\\ &= \alpha T_2(T_1(u)) + T_2(T_1(v))\\ &= \alpha(T_2 \circ T_1)(u) + (T_2 \circ T_1)(v)\end{aligned}$$

Q.E.D

21.17 Proof of proposition of basis transformed by isomorphism in section 14.13.3

21.17.1 i

$0 = \alpha_1 v_1 + \cdots + \alpha_n v_n \Leftrightarrow 0 = T(\alpha_1 v_1 + \cdots + \alpha_n v_n) = \alpha_1 T(v_1) + \cdots + \alpha_n T(v_n)$

21.17.2 ii & iii

($\Rightarrow$)

$w \in W$, $\exists v \in V$ s.t. $T(v) = w$. $v = \sum_{i=1}^{n} \alpha_i v_i$, then $w = \sum_{i=1}^{n} \alpha_i T(v_i)$. So $W = \text{Span}\{T(v_1), T(v_2), \cdots, T(v_n)\}$.

($\Leftarrow$)

$v \in V$. Then $T(v) = w \in W$, $\exists \alpha_1, \alpha_2, \cdots, \alpha_n \in \mathbb{F}$

s.t. $w = \sum_{i=1}^{n} \alpha_i T(v_i) = T(\sum_{i=1}^{n} \alpha_i v_i) = T(\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n)$.

but $w = T(v)$, thus $T(v) - T(\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n) = 0$, $T(v - \sum_{i=1}^{n} \alpha_i v_i) = 0$.

then $v = \sum_{i=1}^{n} \alpha_i v_i$. So $V = \text{Span}\{v_1, v_2, \cdots, v_n\}$.

Q.E.D.

21.18 Proof of proposition dimension of isomorphic spaces in section 14.13.4

Using (iii) in the former proposition, $\dim_{\mathbb{F}}(V) = \dim_{\mathbb{F}}(W)$.

Q.E.D.

21.19 Proof of the theorem in section 14.15.1

$\dim_{\mathbb{F}}(V) = \dim_{\mathbb{F}}(W) = n$, then $V \cong_{\mathbb{F}} \mathbb{F}^n$ and $\mathbb{F}^n \cong_{\mathbb{F}} W$, and $V \cong_{\mathbb{F}} W$.

Q.E.D.

21.20 Proof of lemma of invertible matrices in section 6.12.2

Suppose $BA = I$ and $AC = I$, then $B = BI = B(AC) = (BA)C = IC = C$, so $B = C$.

Q.E.D.

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